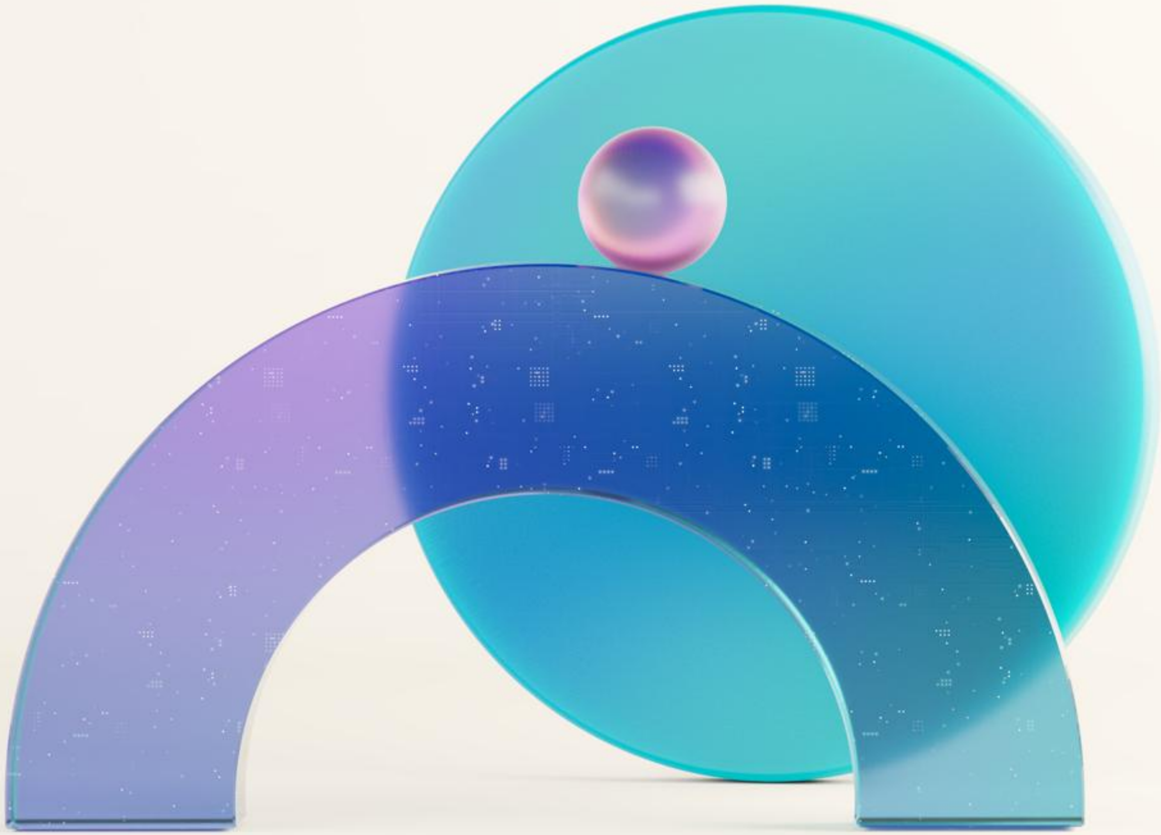
Microsoft



Architecture Design & Review

Microsoft Fabric

Agenda

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Our delivery approach

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The necessity of
Real-Time Intelligence

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Fabric RTI Components

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Architectures
Design & Patterns

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Business Discovery

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Technical Discovery

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Analysis & Planning

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Executive Presentation

Our delivery approach

Objectives of the "Architecture Design & Review"

Transferring you the knowledge about the tech stack of Real-Time Intelligence and different Architecture Design and Patterns.

Understand your environment from Business and Technical perspectives and assess gaps from the Best Practices

Provide you with clarity about the value of Real-Time Intelligence; propose solutions to your challenges; inspire you with ideas of use cases.

Expectations From You

- Ask questions, participate, engage
- Help us understand your business objectives
- Guide us through your current technical environment
- Provide constructive feedback or ideas on improving the delivery

Engagement Timeline

Day 1 | Architecture

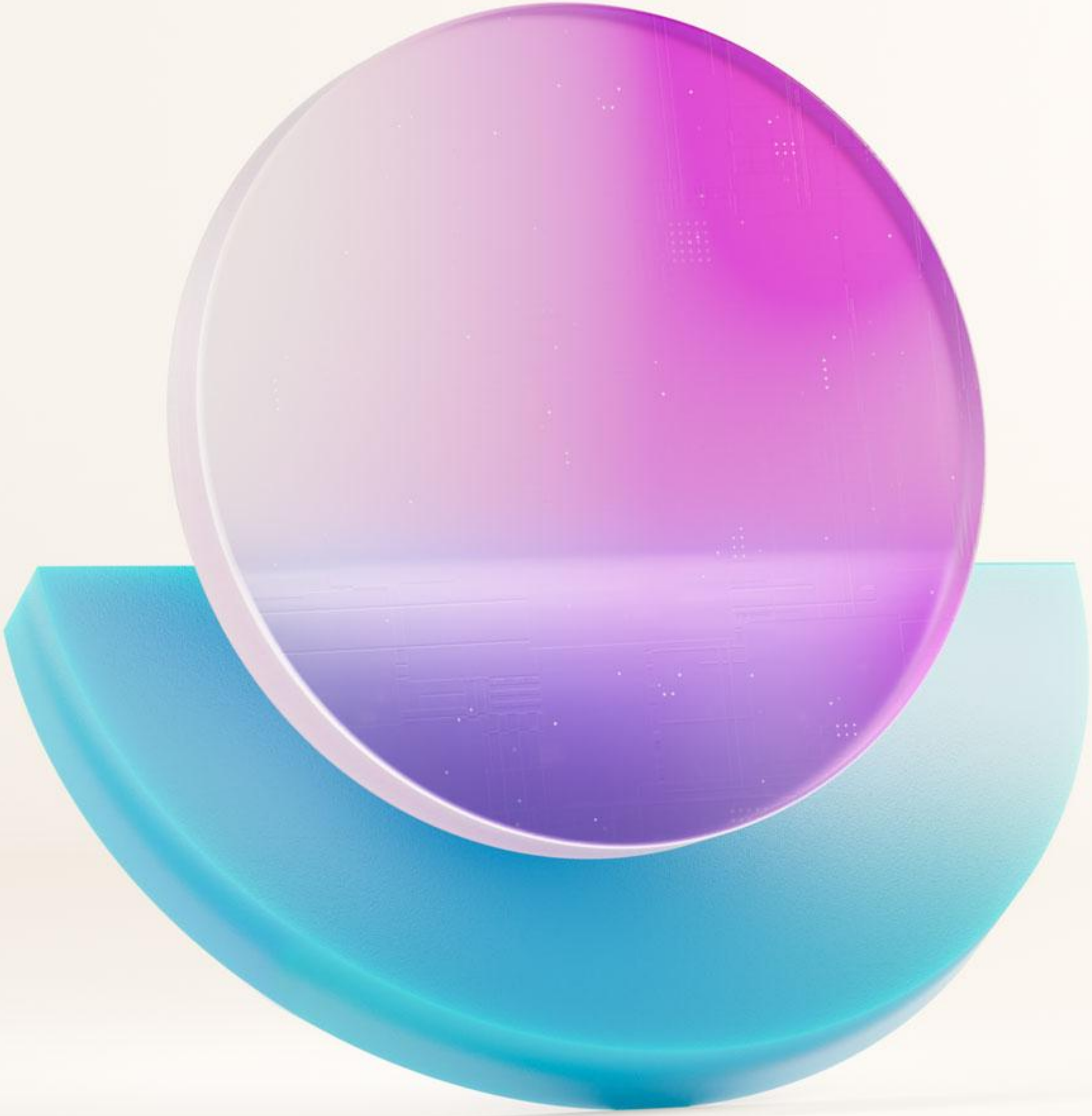
- Kick-Off
- Customer Stories
- Fabric RTI Components
- Architectures Design & Patterns

Day 2 | Discovery

- Business Discovery
- Technical Discovery
- Analysis of Needs

Day 3 | Planning

- Findings
- Best Practices
- Action Plan
- Proposed Architecture
- Offerings
- Engagement Report
- Engagement Presentation
- Close-Out



Customer Stories

Customer Success Story

Who are they?

Customeris a leading UK-based supermarket chain specialising in frozen foods, with over 1,000 stores. It plays a critical role in the retail sector, serving millions of customers weekly and managing complex supply chains. It is known for its competitive pricing and strong brand loyalty.

What’s the Challenge?

Iceland Foods faced a significant challenge in adapting to rapidly changing consumer behaviour and market conditions. The retail industry is highly competitive, and customer expectations for personalised offers, real-time stock availability, and seamless shopping experiences have never been higher. Iceland’s traditional data systems were batch-oriented, meaning insights were delayed by hours or even days. This lag hindered the company’s ability to respond to sudden demand spikes, optimise promotions, and manage inventory efficiently. For example, during seasonal peaks or unexpected events, the inability to access real-time data often resulted in stockouts or overstocking, leading to lost sales and increased waste. Additionally, marketing teams struggled to measure campaign performance in real time, making it difficult to adjust strategies dynamically. These inefficiencies not only impacted revenue but also customer satisfaction, as shoppers increasingly expect instant responses and personalised experiences.

What’s the Solution?

To overcome these challenges, Iceland Foods implemented Microsoft Fabric Real-Time Intelligence (RTI). This solution unified data streams from multiple sources, including point-of-sale systems, online transactions, and supply chain sensors, into a single, real-time analytics platform. Using RTI, Iceland could ingest and process millions of events per second, enabling near-instant insights into sales trends, inventory levels, and customer behaviour. For instance, when a promotion was launched, the marketing team could monitor its performance in real time and adjust offers or pricing dynamically to maximise ROI. Similarly, store managers gained dashboards that refreshed every few seconds, allowing them to make immediate decisions on stock replenishment and staffing. The integration with predictive analytics further enhanced decision-making by forecasting demand patterns based on live data. As a result, Iceland achieved significant improvements in operational efficiency, reduced waste, and enhanced customer satisfaction through timely and personalised offers. This transformation positioned Iceland as a leader in leveraging real-time intelligence in retail.

Customer Success Story

Who are they?

Customer is the world's largest provider of flexible workspace solutions, operating well-known brands such as Regus, Spaces, and HQ. With a presence in over 120 countries and thousands of locations, IWG serves millions of professionals and businesses, playing a pivotal role in shaping the future of work.

What's the Challenge?

The global shift towards hybrid work models has created unprecedented demand for flexible office solutions. While this trend presented a significant growth opportunity for IWG, it also introduced complex operational and marketing challenges. The company needed to manage a vast network of workspaces, optimise occupancy rates, and deliver personalised experiences to a diverse customer base—all while maintaining profitability in a highly competitive market.

One of the most pressing issues was the lack of real-time visibility into marketing campaign performance and sales activities across multiple regions. IWG's traditional reporting systems relied heavily on batch processing, which meant that data was often delayed by 24 hours or more. This delay severely limited the company's ability to make timely decisions. For example, if a digital marketing campaign underperformed in a specific region, the marketing team would only discover this after the campaign had run its course, resulting in wasted ad spend and missed revenue opportunities.

Fraud detection was another critical challenge. With thousands of transactions occurring daily across multiple geographies, IWG faced a growing risk of fraudulent activities, such as fake bookings or misuse of promotional offers. The existing fraud detection mechanisms were reactive rather than proactive, often identifying issues only after financial losses had occurred.

Operational inefficiencies extended to sales performance monitoring. Sales teams lacked access to up-to-the-minute data on lead conversions, pipeline health, and regional performance metrics. This lack of visibility hindered their ability to prioritise high-value opportunities and respond quickly to market dynamics. As a result, IWG struggled to maintain consistent revenue growth and customer satisfaction in an increasingly competitive landscape.

In summary, the absence of real-time insights created a cascade of challenges for IWG: ineffective marketing spend, increased fraud risk, and suboptimal sales performance. These issues collectively threatened the company's ability to capitalise on the booming demand for flexible workspaces and maintain its leadership position in the market.

What's the Solution?

To address these challenges, IWG implemented **Microsoft Fabric Real-Time Intelligence (RTI)**, a comprehensive solution designed to deliver actionable insights at scale and speed. The deployment began with integrating data streams from multiple sources, including marketing platforms, CRM systems, and financial transaction logs, into a unified real-time analytics environment powered by Fabric RTI.

One of the key benefits of this solution was its ability to provide instant visibility into marketing campaign performance. Marketing teams could now monitor key metrics such as click-through rates, conversion rates, and cost per acquisition in real time. This capability enabled them to make dynamic adjustments to campaigns—shifting budgets to high-performing channels, pausing underperforming ads, and testing new creative strategies on the fly. As a result, IWG significantly improved its marketing ROI and reduced wasted spend.

Fraud detection also underwent a dramatic transformation. By leveraging RTI's advanced analytics and machine learning capabilities, IWG implemented real-time anomaly detection models that flagged suspicious transactions within seconds. This proactive approach allowed the company to prevent fraudulent activities before they resulted in financial losses, thereby safeguarding revenue and enhancing customer trust.

Sales teams benefited from live dashboards that provided a holistic view of pipeline health, lead conversion rates, and regional performance metrics. These dashboards refreshed every few seconds, empowering sales managers to make data-driven decisions in real time. For example, if a particular region experienced a sudden drop in conversions, the sales team could investigate and address the issue immediately, rather than waiting for end-of-day reports.

The implementation of Microsoft Fabric RTI not only resolved IWG's immediate challenges but also laid the foundation for long-term innovation. By harnessing the power of real-time data, IWG transformed its operations from reactive to proactive, enabling the company to deliver superior customer experiences, optimise resource allocation, and maintain its competitive edge in the rapidly evolving flexible workspace market.

A major telecommunications Provider in New Zealand

Who are they?

A leading telecommunications provider serving millions of customers across New Zealand. It offers mobile, broadband, and enterprise solutions, playing a critical role in the country's digital infrastructure and connectivity.

What's the Challenge?

As a major telecom operator, this customer operates in an environment where customer expectations for speed, reliability, and personalisation are extremely high. The company's success depends on its ability to deliver seamless connectivity and exceptional customer experiences. However, this customer faced significant challenges due to limitations in its existing analytics infrastructure.

The core issue was the lack of real-time visibility into network performance, customer interactions, and operational metrics. Dashboards used by customer service and operations teams refreshed only once per minute or longer. While this might seem fast in some industries, in telecommunications—where thousands of events occur every second—such delays can have serious consequences. For example, when a network outage or degradation occurred, the delay in detecting and diagnosing the issue often resulted in prolonged service disruptions, negatively impacting customer satisfaction and brand reputation.

Customer experience management was another area of concern. this customer aimed to provide personalised offers and proactive support to its customers, but the absence of real-time insights made this goal difficult to achieve. Marketing and customer care teams relied on historical data, which meant that offers were often generic and reactive rather than timely and relevant. This lack of agility limited the company's ability to differentiate itself in a highly competitive market.

Operational inefficiencies compounded these challenges. Teams responsible for monitoring network health and customer interactions had to work with fragmented data sources and delayed reports. This not only slowed down decision-making but also increased the risk of errors. For instance, when a surge in customer complaints occurred, it could take several minutes—or even hours—to identify the root cause and deploy corrective measures. In an industry where every second counts, these delays translated into lost revenue, increased churn, and diminished customer trust.

In summary, this customer's reliance on near-real-time (but not truly real-time) analytics created a reactive operating model. The company needed a solution that could deliver instant insights, enable proactive decision-making, and support its vision of becoming a customer-centric, data-driven organisation.

What's the Solution?

To overcome these challenges, this customer implemented **Microsoft Fabric Real-Time Intelligence (RTI)**, a next-generation analytics solution designed to process and analyse data streams at scale and speed. The deployment focused on three key areas: network performance monitoring, customer experience optimisation, and operational efficiency.

The first step was to integrate data from multiple sources—including network sensors, customer interaction logs, and CRM systems—into a unified real-time analytics platform powered by Fabric RTI. This integration enabled this customer to ingest and process millions of events per second, providing a single source of truth for all operational and customer-related data.

One of the most impactful outcomes was the dramatic improvement in dashboard refresh rates. Previously, dashboards updated every 60 seconds; with RTI, they now refresh every 10 seconds—a sixfold improvement. This enhancement allowed network operations teams to detect and respond to issues almost instantly, significantly reducing downtime and improving service reliability. For example, when a network anomaly occurred, the system could trigger alerts in real time, enabling engineers to take corrective action before customers were affected.

Customer experience also saw a major transformation. By leveraging real-time insights, this customer could deliver personalised offers and proactive support based on live customer behaviour. For instance, if a customer experienced a service disruption, the system could automatically trigger a goodwill offer or escalate the issue to a support agent within seconds. This level of responsiveness not only improved customer satisfaction but also strengthened brand loyalty.

Operational efficiency improved across the board. Teams gained access to live dashboards that provided a holistic view of key metrics, enabling faster and more informed decision-making. Predictive analytics models built on top of RTI further enhanced this capability by forecasting potential issues before they occurred, allowing this customer to move from a reactive to a proactive operating model.

The implementation of Microsoft Fabric RTI positioned this customer as a leader in digital innovation within the telecommunications sector. By harnessing the power of real-time data, the company achieved significant improvements in service reliability, customer satisfaction, and operational agility—key differentiators in an increasingly competitive market.

One of the largest US supply chain services companies

Who are they?

This customer is one of the largest supply chain services companies in the United States. It provides grocery and foodservice supply chain solutions to major retailers, restaurants, and convenience stores, operating a vast distribution network that moves billions of dollars' worth of goods annually.

What's the Challenge?

The logistics and distribution industry is under constant pressure to deliver goods faster, more efficiently, and at lower costs. For this customer, which manages thousands of daily deliveries across a nationwide network, operational efficiency is critical to maintaining profitability and customer satisfaction. However, the company faced significant challenges in optimising its transportation and distribution processes due to limitations in its existing data systems.

One of the most pressing issues was route optimisation. this customer's legacy systems relied on static routing plans that were updated periodically, often based on historical data rather than real-time conditions. This approach failed to account for dynamic variables such as traffic congestion, weather disruptions, and last-minute order changes. As a result, delivery routes were frequently suboptimal, leading to increased fuel consumption, longer delivery times, and higher operational costs.

Another major challenge was the lack of real-time visibility into fleet operations. Dispatchers and logistics managers had limited ability to monitor vehicle locations, delivery progress, and driver performance in real time. This lack of transparency made it difficult to respond quickly to disruptions, such as vehicle breakdowns or unexpected delays. When issues arose, corrective actions were often reactive and delayed, resulting in missed delivery windows and dissatisfied customers.

Inventory management was also impacted by these inefficiencies. Without real-time data on delivery status, warehouse teams struggled to coordinate replenishment schedules accurately. This misalignment often led to bottlenecks in the supply chain, with some locations experiencing stockouts while others held excess inventory. These imbalances not only increased costs but also eroded service levels for this customer's retail and foodservice customers.

In addition to operational inefficiencies, this customer faced growing pressure to reduce its environmental footprint. Suboptimal routing and empty truck runs contributed to unnecessary fuel consumption and carbon emissions, undermining the company's sustainability goals. With increasing regulatory scrutiny and customer expectations for greener supply chains, this customer recognised the urgent need for a more agile, data-driven approach to logistics management.

In summary, this customer's reliance on delayed and fragmented data created a reactive operating model that was ill-suited to the demands of modern supply chain management. The company needed a solution that could deliver real-time insights, enable dynamic decision-making, and support its objectives for efficiency, customer satisfaction, and sustainability.

What's the Solution?

To address these challenges, this customer implemented Microsoft Fabric Real-Time Intelligence (RTI), a powerful analytics solution designed to process and analyse data streams at scale. The deployment focused on transforming this customer's logistics operations from reactive to proactive by providing real-time visibility and predictive insights across the entire supply chain.

The first step was to integrate data from multiple sources, including GPS-enabled telematics systems, traffic and weather feeds, warehouse management systems, and order processing platforms, into a unified real-time analytics environment powered by Fabric RTI. This integration enabled this customer to ingest and process millions of data points per second, creating a single source of truth for all operational decisions.

One of the most impactful outcomes was the implementation of dynamic route optimisation. Using RTI's advanced analytics and machine learning capabilities, this customer developed algorithms that continuously analysed live data on traffic conditions, weather patterns, and delivery priorities. These algorithms automatically generated optimised routes in real time, reducing empty miles and improving delivery efficiency. As a result, this customer achieved significant reductions in fuel consumption, transportation costs, and carbon emissions.

Real-time dashboards provided dispatchers and logistics managers with instant visibility into fleet operations. They could monitor vehicle locations, delivery progress, and driver performance in real time, enabling them to respond immediately to disruptions. For example, if a vehicle experienced a breakdown, the system could automatically reroute nearby trucks to cover the affected deliveries, minimising delays and maintaining service levels.

Inventory management also benefited from real-time insights. Warehouse teams could track delivery status in real time, allowing them to coordinate replenishment schedules more accurately and avoid stock imbalances. This improvement enhanced overall supply chain efficiency and reduced carrying costs.

Beyond operational gains, the implementation of Microsoft Fabric RTI supported this customer's sustainability objectives. By reducing empty truck runs and optimising routes, the company significantly lowered its carbon footprint, aligning with both regulatory requirements and customer expectations for environmentally responsible practices.

The adoption of real-time intelligence marked a transformative shift for this customer. The company moved from a static, reactive operating model to a dynamic, data-driven approach that delivered measurable improvements in efficiency, cost savings, and customer satisfaction. This transformation not only strengthened this customer's competitive position but also reinforced its commitment to innovation and sustainability in the supply chain industry.

A large hospitality company in Europe

Who are they?

This customer is a one of the largest hospitality companies in Europe, managing 36 hotels and 15 resorts along the Adriatic coast. It is a key player in the European tourism industry, renowned for delivering premium guest experiences and driving innovation in hospitality services.

What's the Challenge?

The hospitality industry thrives on delivering exceptional guest experiences, and in today's digital-first world, this requires real-time insights into guest behaviour, preferences, and operational performance. this customer, despite its strong market position, faced significant challenges in achieving this level of agility due to limitations in its existing data infrastructure.

One of the primary issues was the lack of real-time visibility into guest interactions across multiple touchpoints, including reservations, check-ins, dining, and recreational activities. Valamar's legacy systems relied on batch processing, which meant that data was often delayed by several hours or even a full day. This delay hindered the company's ability to personalise guest experiences in the moment. For example, if a guest expressed interest in a spa service during check-in, the marketing team could not act on this information immediately, resulting in missed upsell opportunities.

Operational efficiency was another area of concern. Managing 36 hotels and 15 resorts requires precise coordination of resources such as housekeeping, maintenance, and food and beverage services. Without real-time data, Valamar struggled to optimise staffing levels and resource allocation. This often led to inefficiencies such as overstaffing during low-demand periods or understaffing during peak times, negatively impacting both costs and service quality.

The lack of real-time insights also affected Valamar's ability to respond to guest feedback promptly. While the company collected feedback through various channels, the delayed processing of this data meant that issues were often addressed after the guest had checked out, reducing the effectiveness of service recovery efforts and impacting guest satisfaction scores.

In addition to these operational challenges, Valamar faced increasing competition from global hospitality brands and alternative accommodation platforms. To maintain its competitive edge, the company needed to differentiate itself through superior guest experiences and operational excellence. Achieving this required a shift from reactive decision-making based on historical data to proactive strategies driven by real-time intelligence.

In summary, Valamar's reliance on delayed and fragmented data created a reactive operating model that was misaligned with the demands of modern hospitality. The company needed a solution that could provide real-time insights into guest behaviour and operational performance, enabling it to deliver personalised experiences, optimise resource allocation, and enhance overall efficiency.

What's the Solution?

To address these challenges, this customer implemented Microsoft Fabric Real-Time Intelligence (RTI), a cutting-edge analytics solution designed to process and analyse data streams at scale. The deployment focused on creating a unified real-time data platform that integrated information from multiple sources, including property management systems, point-of-sale systems, guest feedback platforms, and IoT-enabled devices across its properties.

One of the most transformative outcomes was the ability to deliver personalised guest experiences in real time. By leveraging RTI, Valamar could analyse guest behaviour and preferences as they occurred, enabling the company to offer tailored recommendations and promotions instantly. For example, if a guest booked a dinner reservation, the system could automatically suggest a wine pairing or a spa appointment, enhancing the overall guest experience and driving incremental revenue.

Operational efficiency also improved significantly. Real-time dashboards provided managers with instant visibility into key metrics such as occupancy rates, housekeeping status, and maintenance requests. This enabled them to allocate resources dynamically based on current demand, reducing costs and improving service quality. For instance, housekeeping teams could prioritise room cleaning based on live check-in and check-out data, ensuring that rooms were ready for incoming guests without unnecessary delays.

The implementation of RTI also enhanced Valamar's ability to respond to guest feedback promptly. Feedback collected through digital channels was processed in real time, allowing staff to address issues while the guest was still on the property. This proactive approach not only improved guest satisfaction but also strengthened brand loyalty.

Beyond immediate operational benefits, the adoption of Microsoft Fabric RTI positioned Valamar as a leader in digital innovation within the hospitality sector. The company gained the ability to forecast demand patterns, optimise pricing strategies, and plan resources more effectively, all based on real-time data. These capabilities enabled Valamar to deliver superior guest experiences, improve operational efficiency, and maintain its competitive edge in an increasingly dynamic market.

The transformation achieved through RTI was not just technological but cultural. Valamar shifted from a reactive, siloed approach to a proactive, data-driven operating model that empowered employees at all levels to make informed decisions in real time. This change reinforced the company's commitment to excellence and innovation, ensuring its continued success in the evolving hospitality landscape.

Let's look at these challenges

Inventory and promotions needed second-by-second adjustments to avoid stockouts and waste.

Detecting fraud or reallocating ad spend after hours or days is too late-the financial loss is already incurred.

Network outages or service degradation cannot wait for batch reports; every second of downtime impacts thousands of customers.

Route optimisation based on yesterday's data is useless when traffic or weather changes in real time.

Guest experience personalisation must happen during the stay, not after checkout.

What do you suggest to solve these needs?

Use Cases Catalog by industry

Finance & Insurance

- ATM Monitoring
- Finance Automation
- Fraud Detection
- Operational Efficiency

Retail

- Inventory Tracking
- Promotions
- Buying Experiences
- Recommender System
- Supply Chain Management

E-Commerce

- User Behavior Monitoring
- Monitor Anomalies (Transactions, Page Loads)
- Personalized Experience

Hospitality

- Stream Booking, Occupancy & Housekeeping
- Real-Time Services Guest Experience

Education

- Monitor Students Attendance
- Real-Time Grading

Healthcare

- Emergency Response
- Remote Patient Monitoring
- Clinical Decision Support
- Surgical Assistance
- Hospital Operations

Energy & Utilities

- Station Monitoring
- Energy Leakage Detection
- Equipment Maintenance & Monitoring
- Failure Monitoring

Logistics

- Delivery Tracking & Routing
- Warehouse Management
- Supply & Demand Operations

Manufacturing

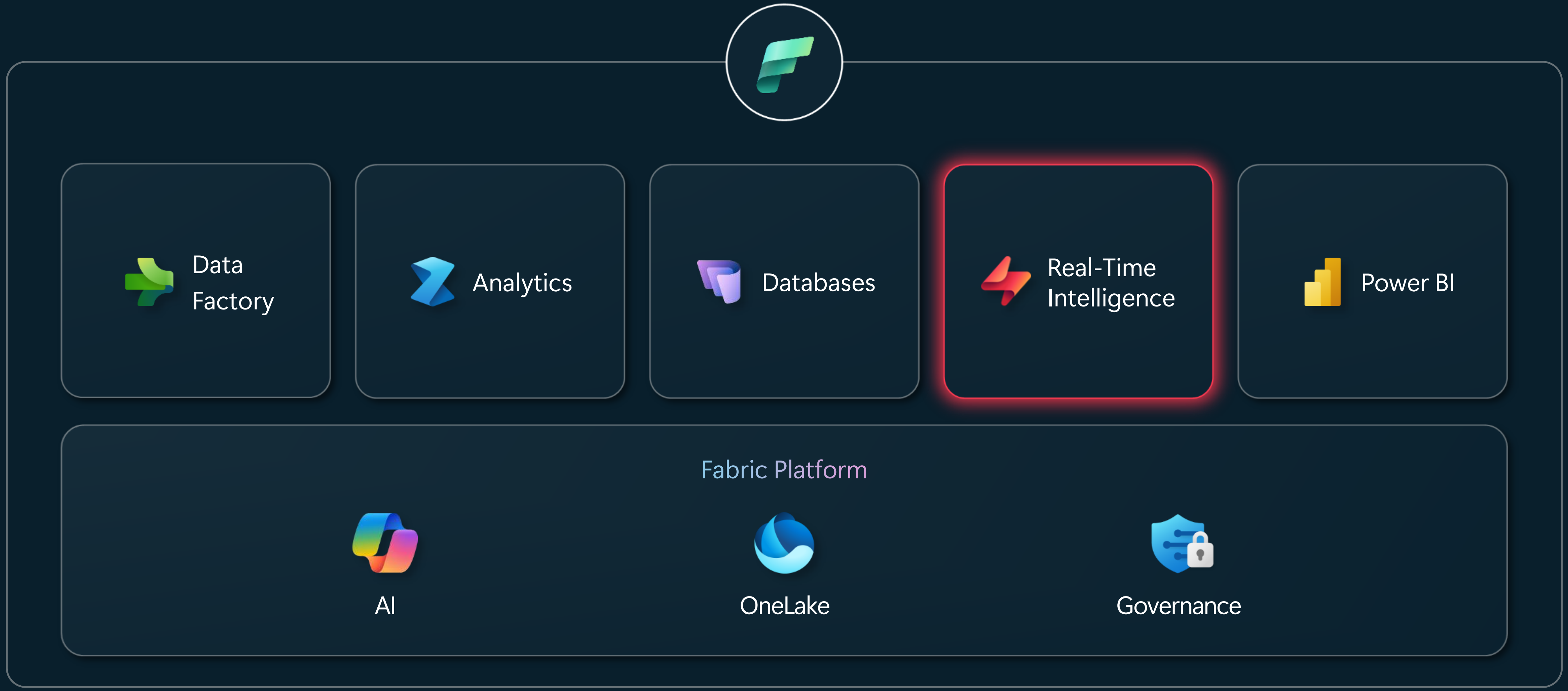
- Quality & Throughput Improvement
- Predictive Maintenance
- Predictive Inventory
- Monitor Unsafe Worker Behaviour

Transportation

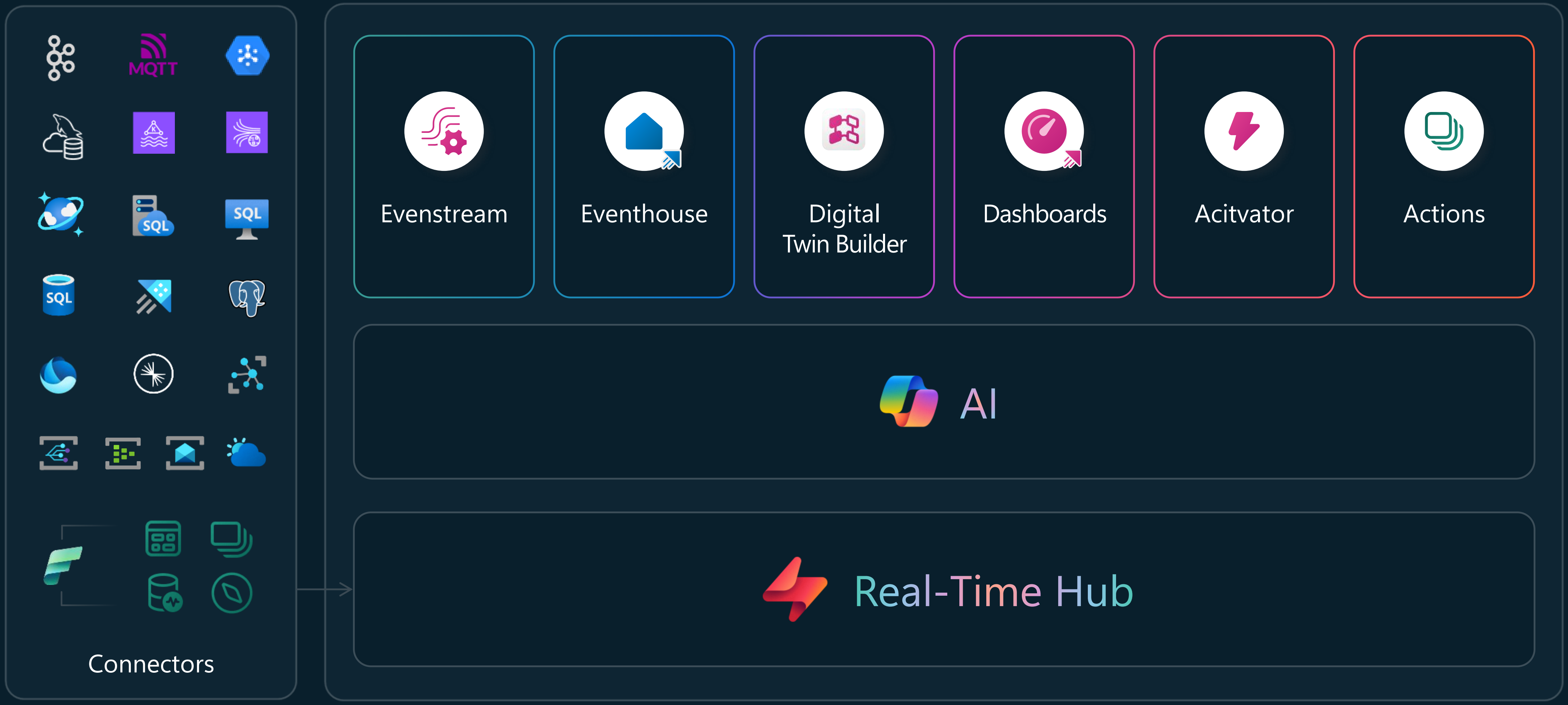
- Route Optimization
- Connected Fleet Applications
- Autonomous Driving
- Manufacturing R&D

Microsoft Fabric

The unified data platform for AI transformation

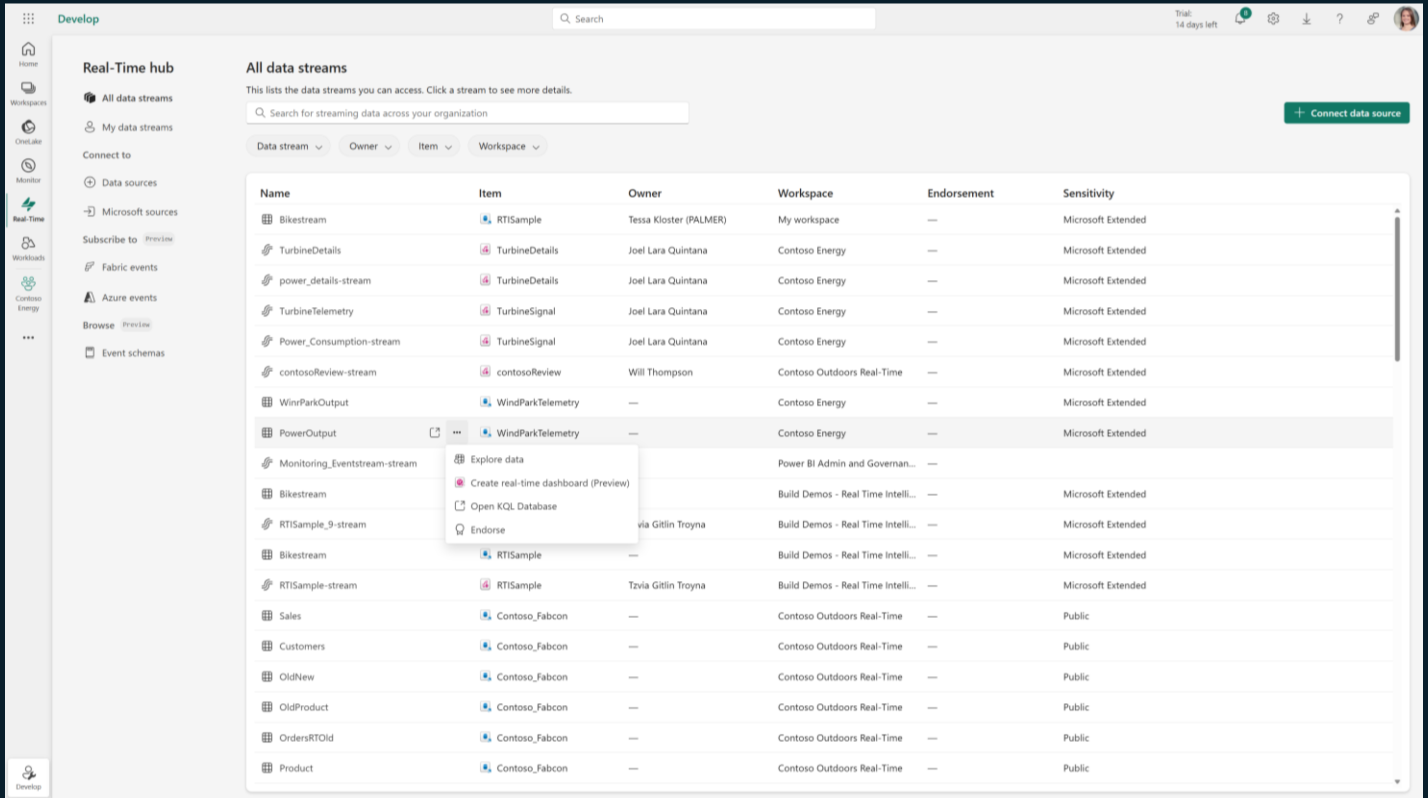


Real-Time Intelligence in Microsoft Fabric



Real-Time Intelligence in Microsoft Fabric

Real-Time Hub



- Find and discover all streaming data in Fabric
- Take common actions on streams/tables
- Explore data and generate Real-Time Dashboards
- Browse existing Microsoft sources with easy connection
- Out-of-box connectors to MSFT and cross-cloud sources
- Create automated workflows and respond to data changes with Fabric Events
- Event schema registry for standardization and data quality

Real-Time Intelligence in Microsoft Fabric

Connectors

Rich set of real-time streaming connectors

Available now

External sources

- Apache Kafka
- Apache Spark
- Apache Flink
- MQTT v5
- Azure SQL DB CDC
- PostgreSQL DB CDC
- MySQL DB CDC
- Azure Cosmos DB CDC
- Azure SQL MI DB CDC
- SQL Server on VM DB CDC
- Azure Data Explorer
- Azure IoT hub

(continued)

- Google Cloud Pub/Sub
- Confluent Cloud for Apache Kafka
- Amazon Kinesis Data Streams
- Amazon Managed Streaming for Apache Kafka
- Solace PubSub+
- Azure Event Hubs
- Azure Service Bus
- Azure Event Grid Namespace
- Real-time weather
- OpenTelemetry
- Telegraf

(continued)

- Splunk Apache Log4j
- FluentBit
- Logstash
- NLog
- Serilog

Fabric & Azure events

- Fabric Workspace Item events
- Fabric Onelake events
- Fabric Job events
- Azure Blob Storage events

Upcoming

External sources

- HTTP
- Mongo DB CDC
- Oracle DB CDC
- Azure blob file stream
- MQTT v3.1.1
- ServiceNow
- OneLake file stream
- Salesforce
- Delta lake CDF
- Eventhouse
- Fabric SQL CES

(continued)

- JDBC
- JIRA

Custom connectors

- SAP Hana

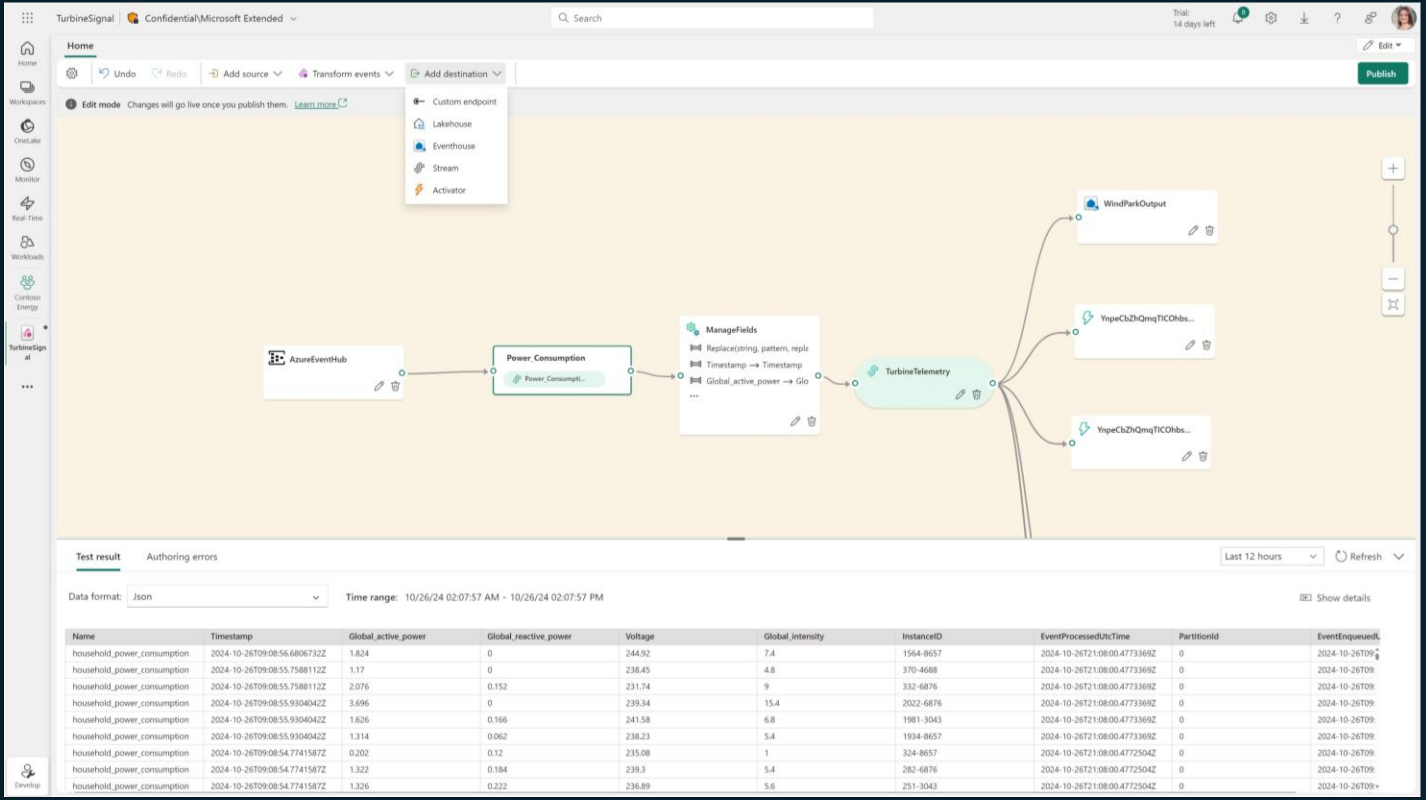
Fabric events

- Anomaly Detection events
- Capacity Utilization events
- Business events

Real-Time Intelligence in Microsoft Fabric

Eventstream

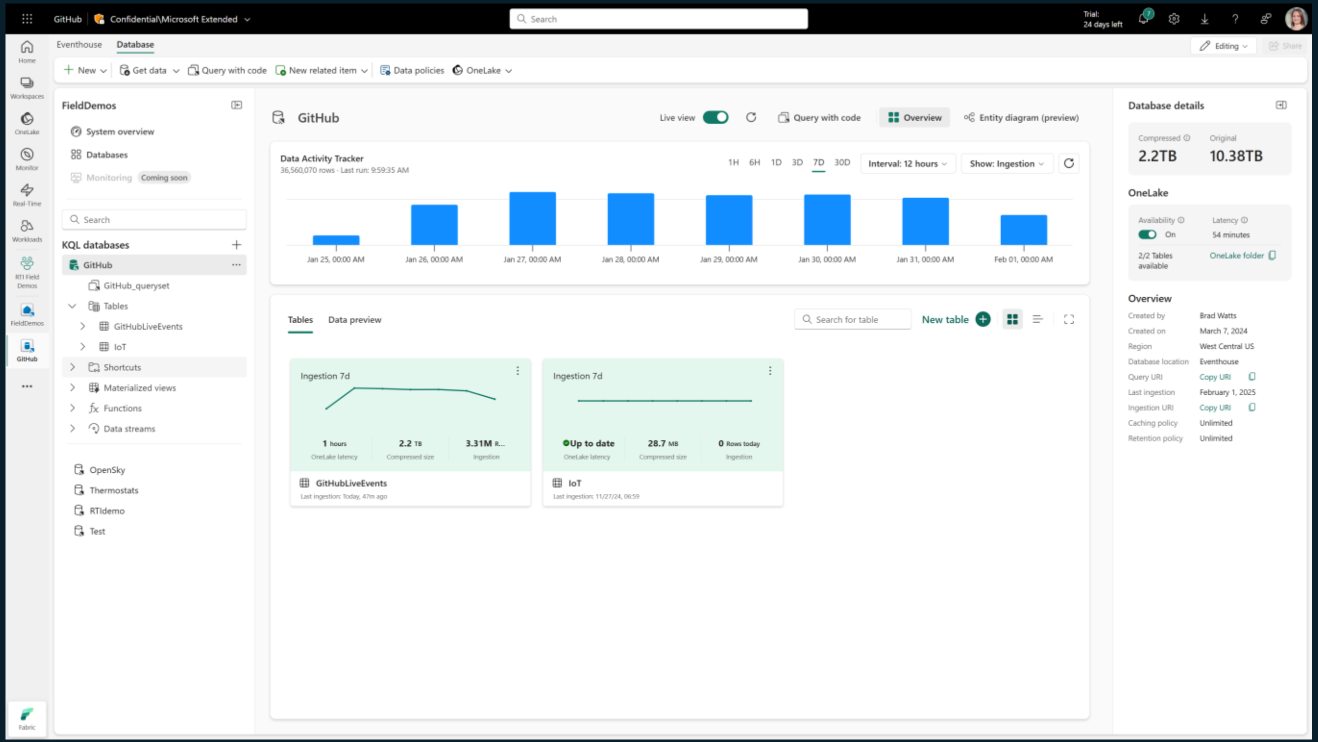
- Connect to wide range of streaming and discrete sources
  - Within MSFT and cross-cloud
  - New support for MQTT, Weather data feeds, Solace PubSub+
- Out-of-box transformations including managing fields, aggregations etc.
  - Including SQL transform
- Content-based routing to multiple destinations including Eventhouse, Activator
- Including CI/CD and Public API support



Real-Time Intelligence in Microsoft Fabric

Eventhouse

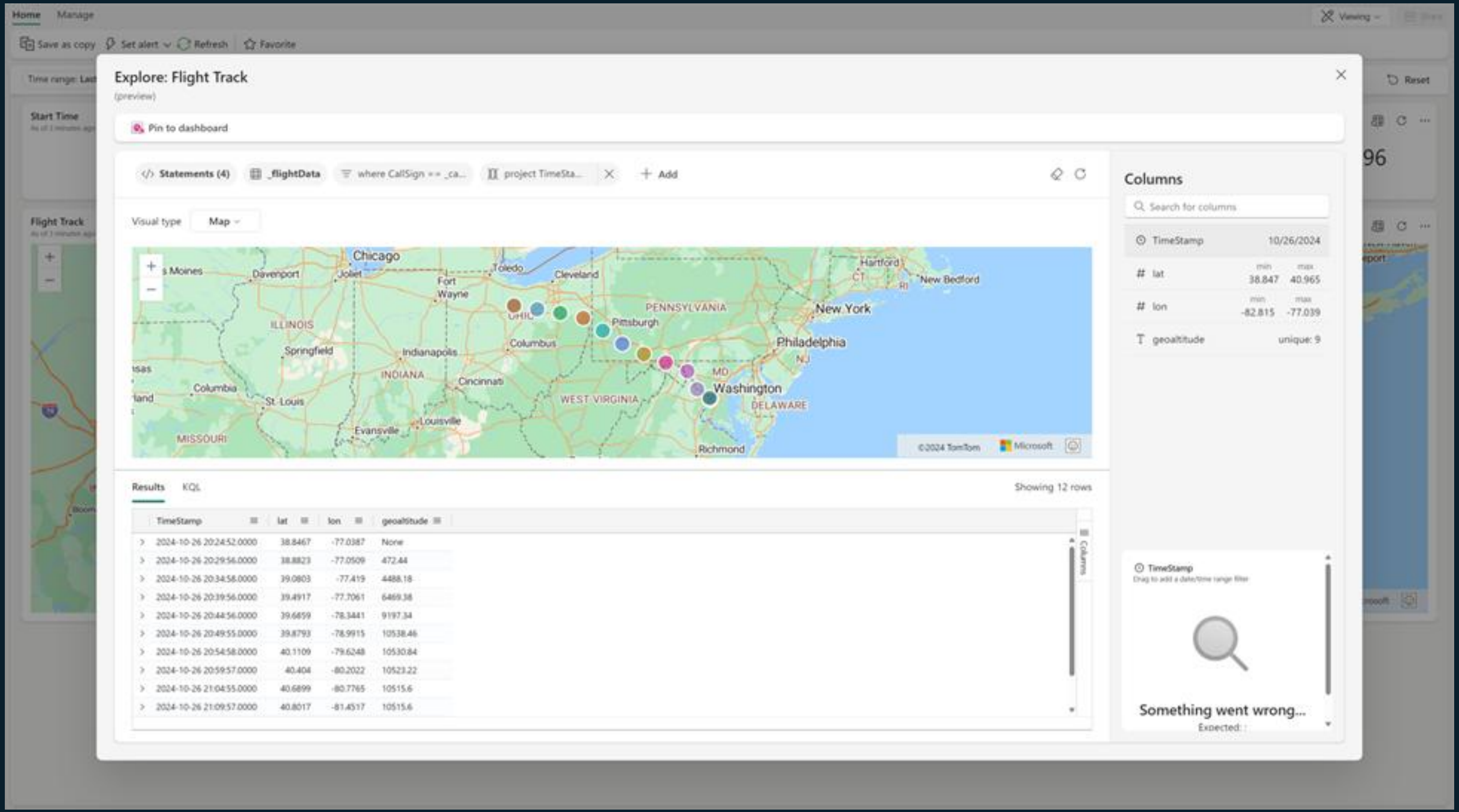
- Specializes in high granularity time & space data streams: activity logs, telemetry and transactions
- Unlimited ingestion volume in low latency
- Human friendly KQL for powerful analytics and no code experiences for business users
- Data lifecycle management: automatic partitioning, retention, tiered storage, GDPR, materialized views
- Query Acceleration (GA) provides at-scale query over data OneLake with same performance with native Eventhouse ingestion



Real-Time Intelligence in Microsoft Fabric

Dashboards

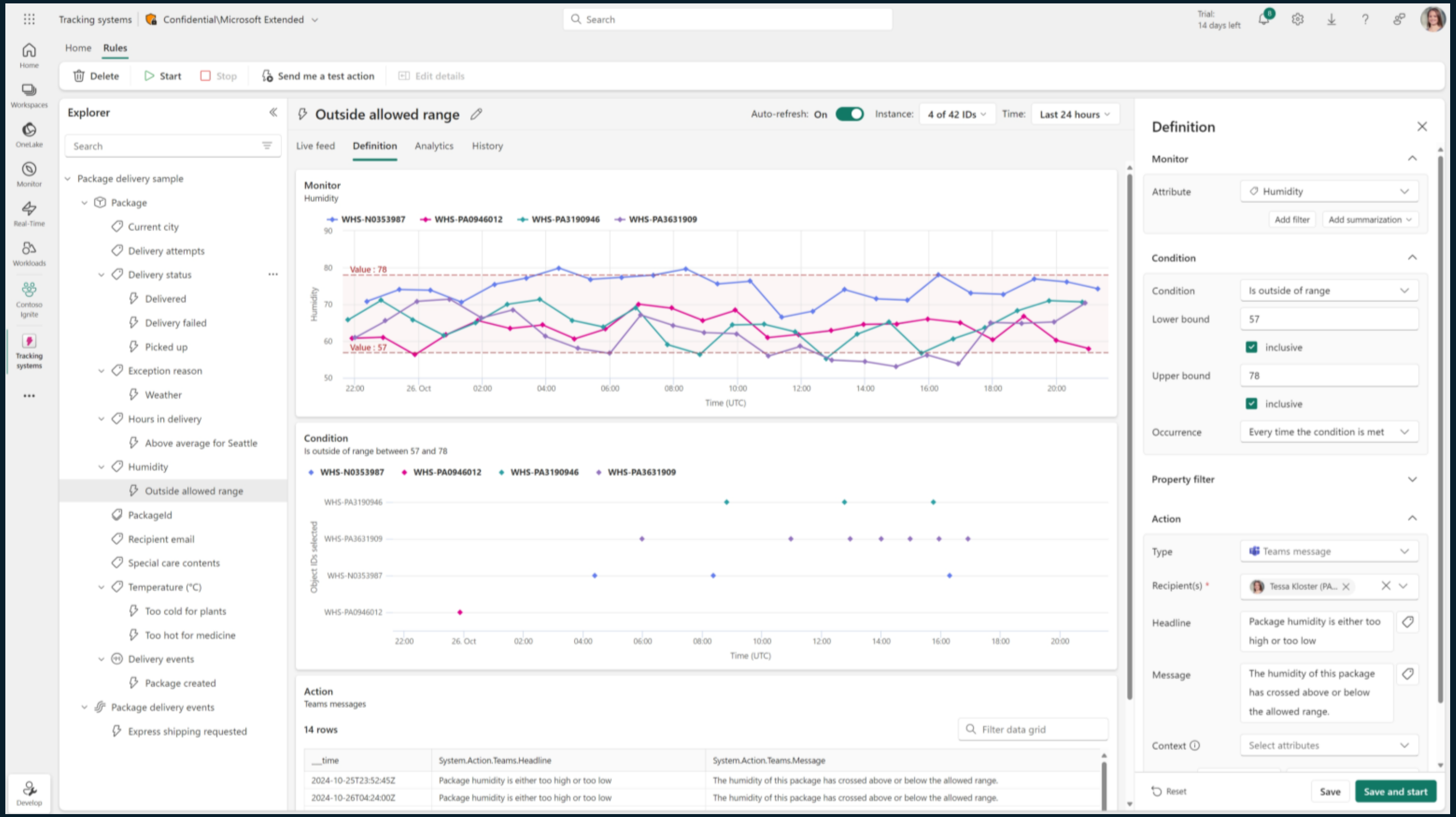
- Operational, live updating visuals and tiles
- No-code data exploration directly from dashboard
- New insights can be pinned to existing or new dashboard
- Share data exploration results using deep link to the data exploration
- Distribution through Power BI organizational apps



Real-Time Intelligence in Microsoft Fabric

Activator

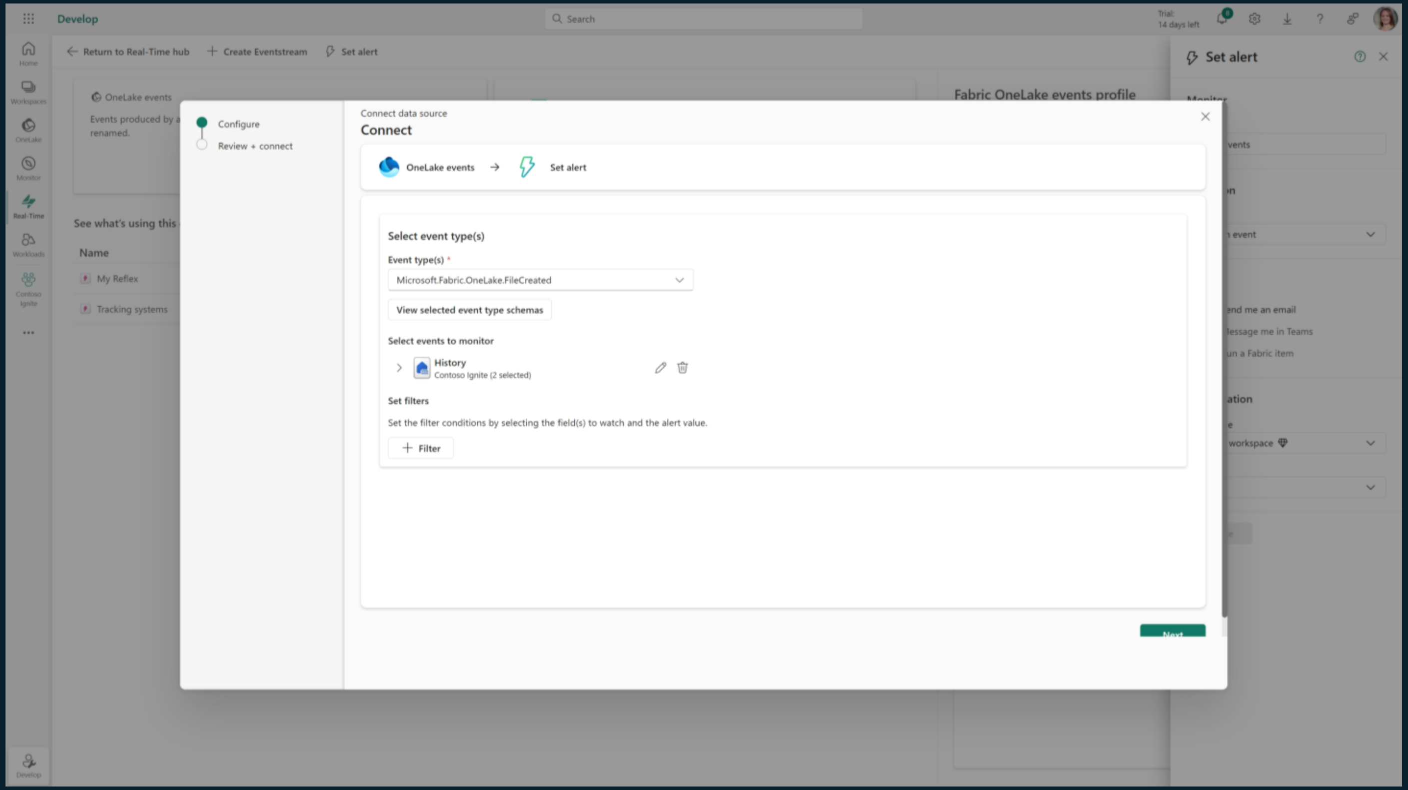
- Take actions when patterns or conditions are detected in changing data
- Monitor data in Eventstreams, Eventhouse, Real-Time Dashboards and Power BI reports
- When the data hits certain thresholds or matches other patterns, automatically alert or trigger workflows



Real-Time Intelligence in Microsoft Fabric

Actions

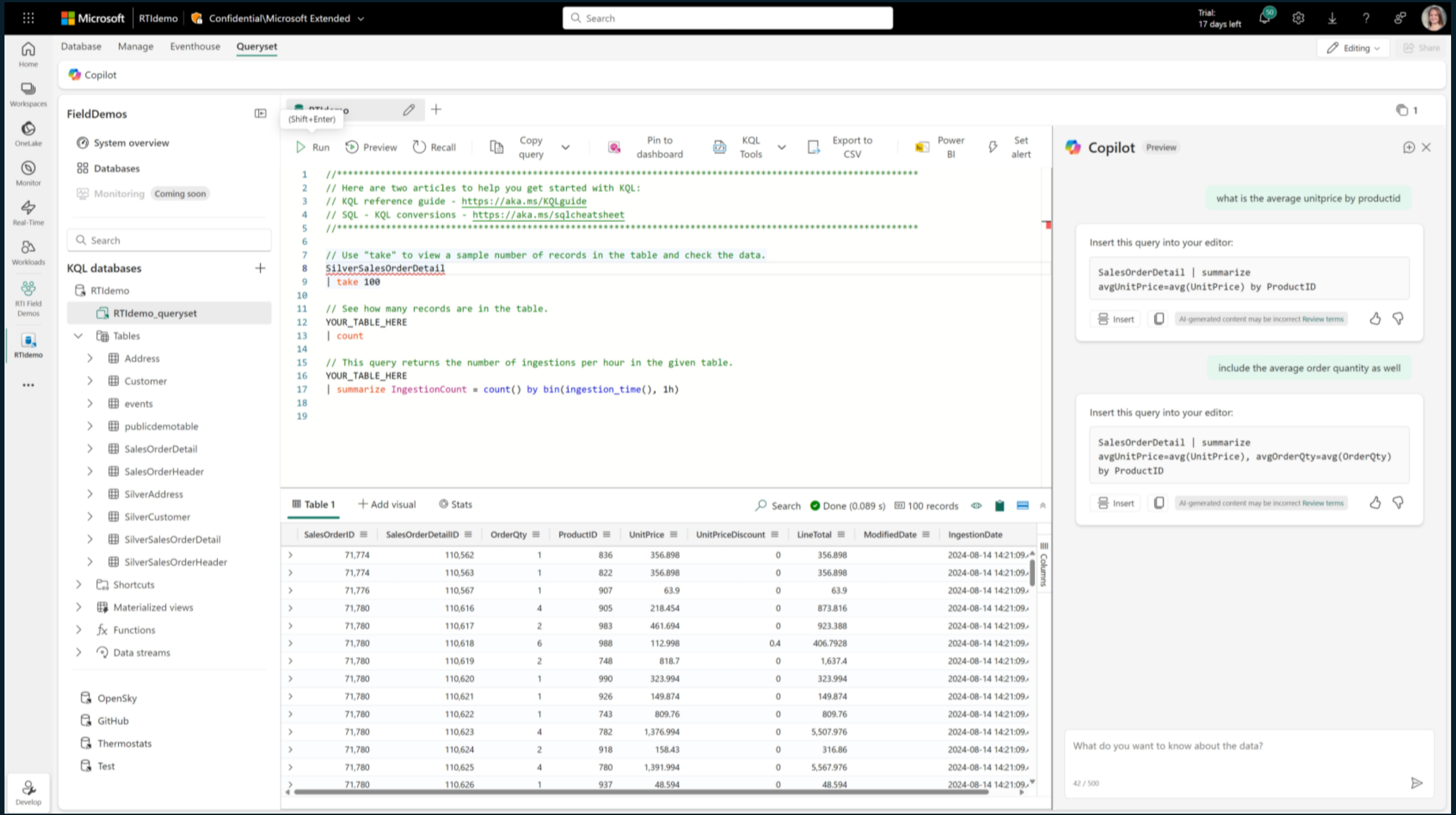
- Simplify the creation of automated workflows and respond to data changes in both Fabric and Azure
- Subscribe to events through Activator action or Eventstream
- Events include
  - System events from Fabric
  - Fabric workspace & job events
  - OneLake events
  - Azure system events from ADLS storage



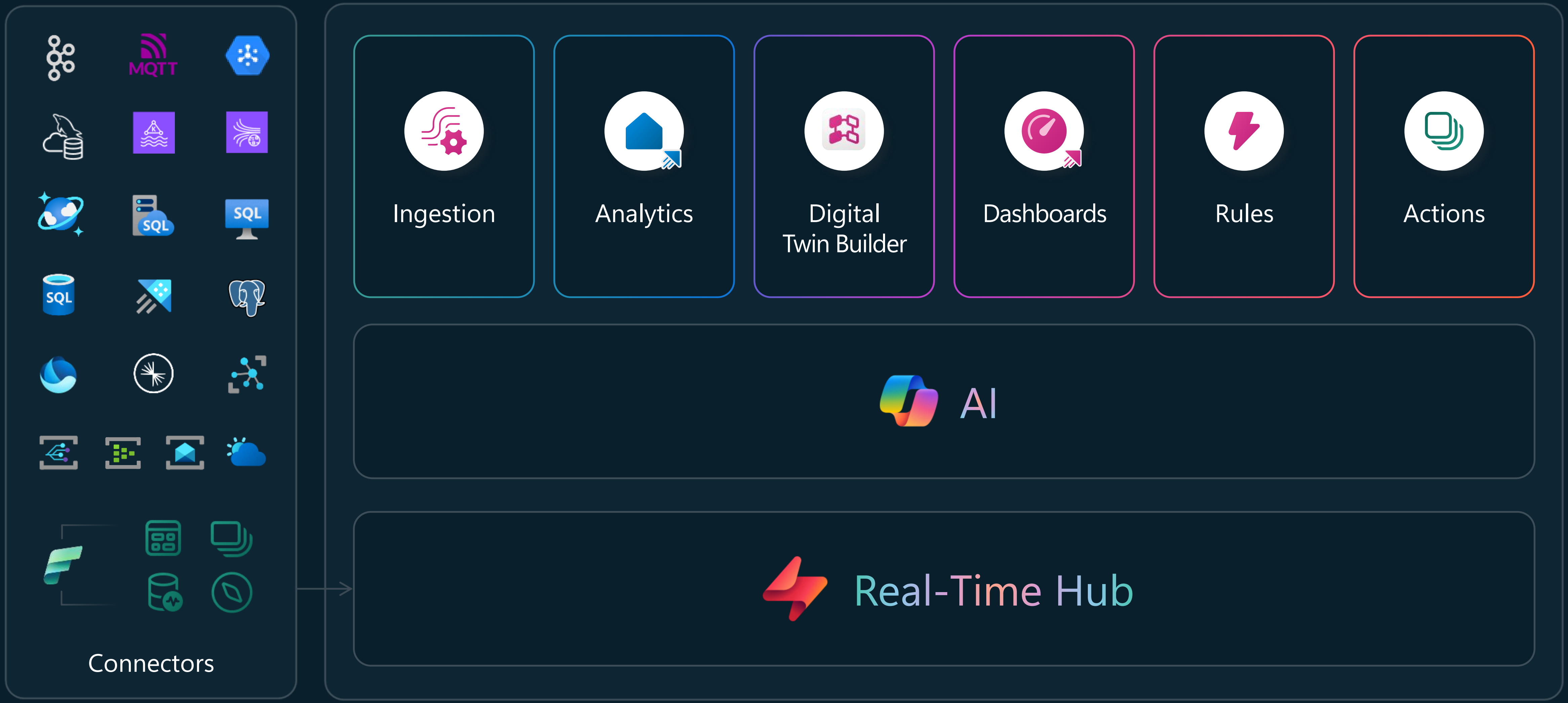
Real-Time Intelligence in Microsoft Fabric

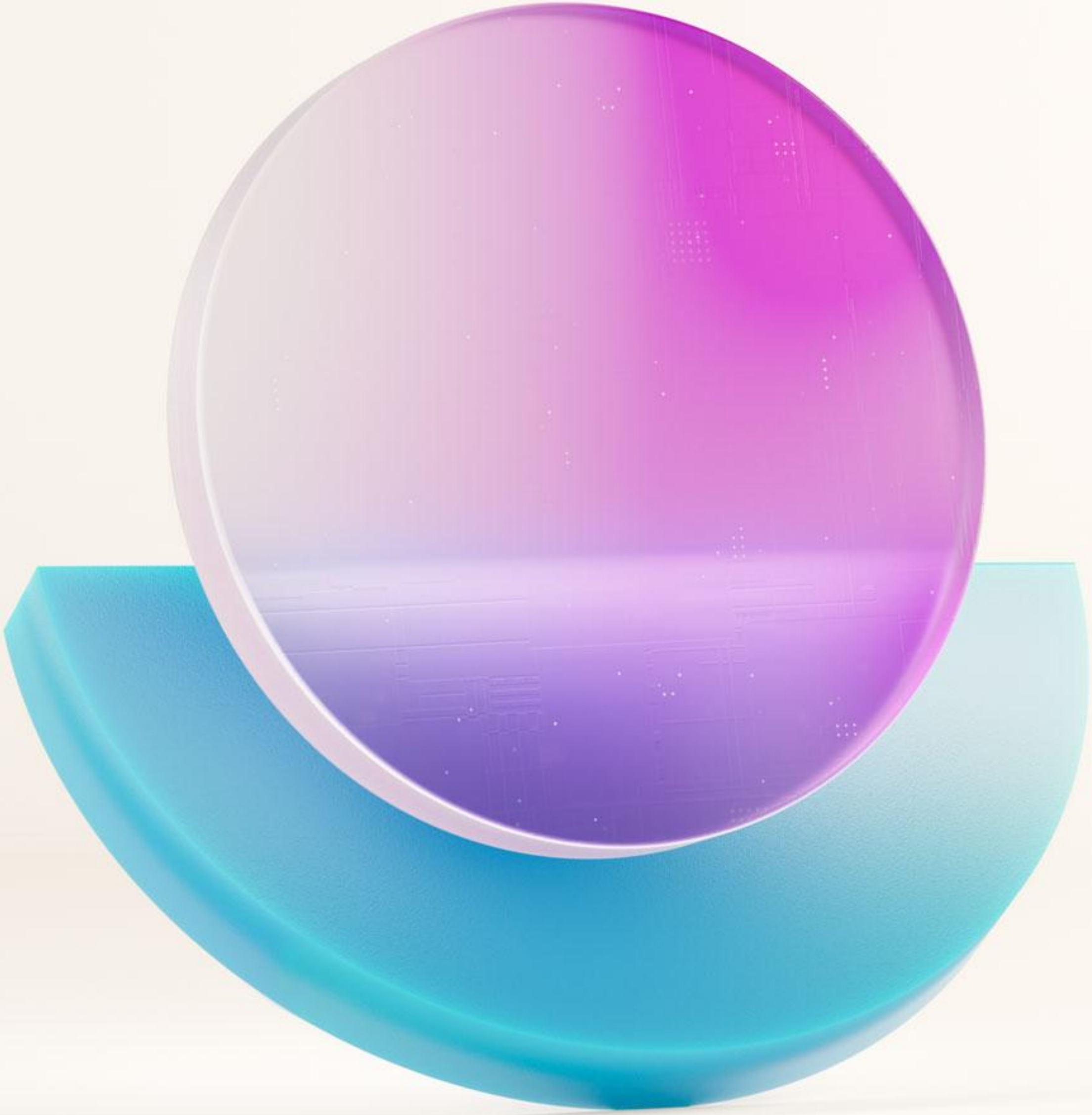
AI

- Generating KQL queries from natural language conversation
- Easy to share / pin results to existing or new dashboards
- Generating Real-Time Dashboards automatically from existing Tables
- AI Skills provides Natural Language Interaction over data in Fabric & Eventhouse



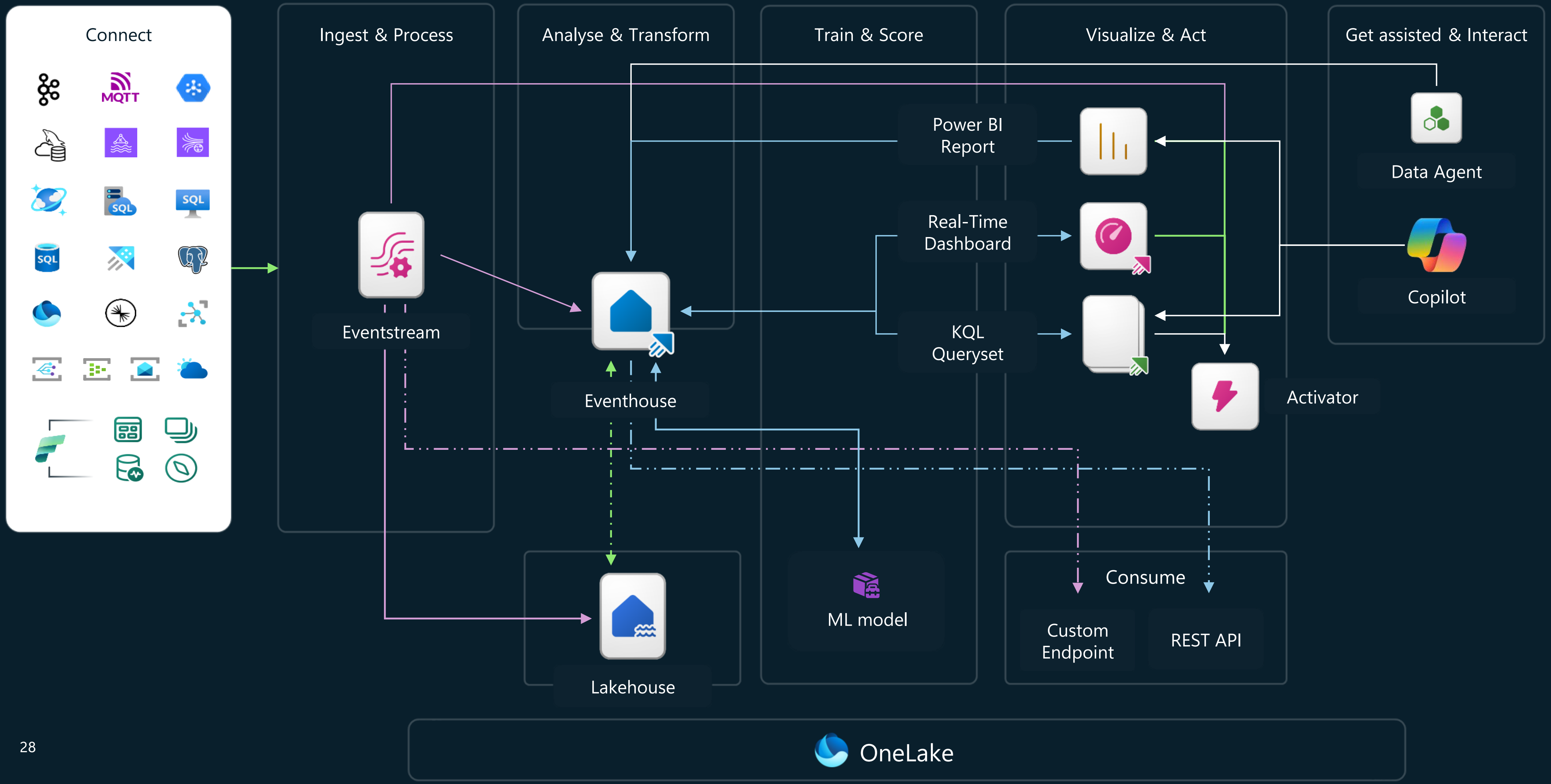
Real-Time Intelligence in Microsoft Fabric





Architecture Design & Patterns

Real-Time Intelligence Architecture Overview



Unify your Batch and Real-Time data

Ingest & process

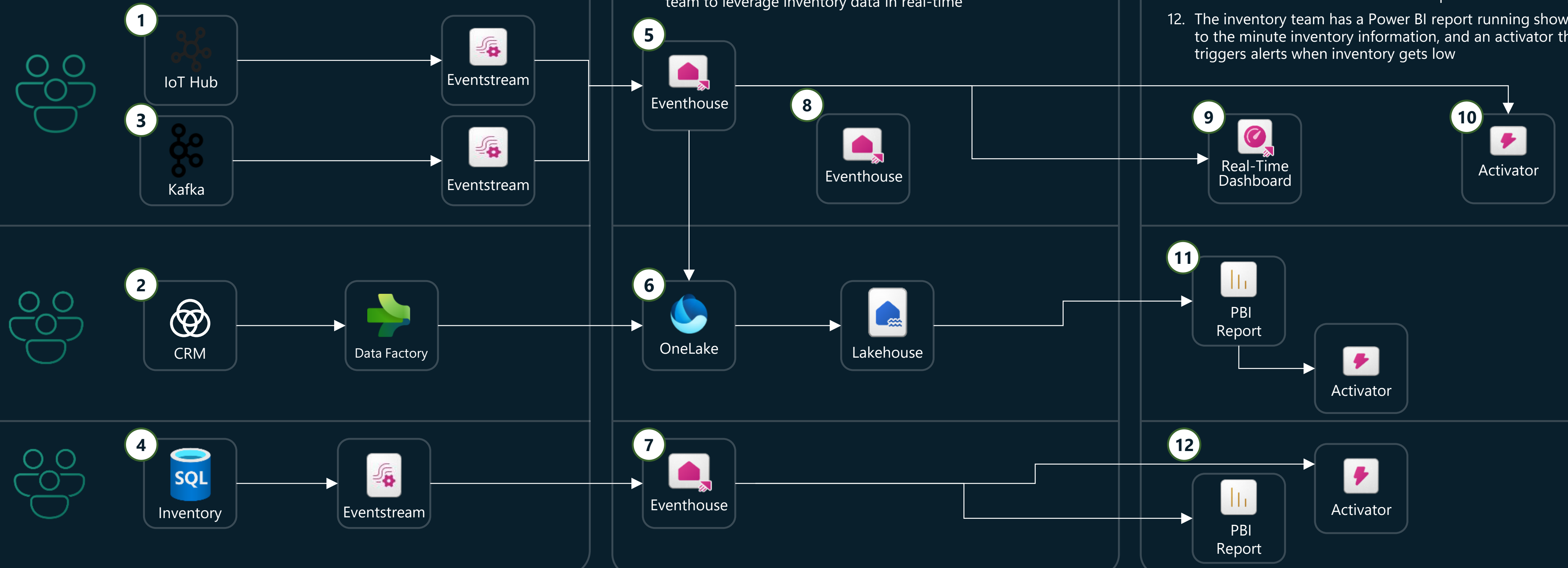
1. A work order team loads vehicle telemetry in real-time into Fabric RTI
2. A finance team maintains financial reporting data in a CRM
3. Work Order data is sent via Kafka topics to an event stream in the work order workspace
4. An inventory team manages the stock across all products in the organization in a SQL Server database

Analyze & transform

5. An eventhouse uses table update policies to load and process data
6. Finance mirrors the data to OneLake and transforms it with CRM data using either SQL or Spark
7. Inventory team loads data to Eventhouse storing up to the minute inventory data
8. An Eventhouse shortcut is created back to the Work Order team to leverage inventory data in real-time

Visualize and Activate

9. A real-time dashboard provides an up-to-date view of the work order system along with a map showing the closest available technician
10. An activator is triggered which creates a dispatch to the nearest available technician
11. A Power BI report runs within Finance that reconciles work orders and CRM data for monthly reporting. Finance creates an activator which creates the invoice upon work order dispatch
12. The inventory team has a Power BI report running showing up to the minute inventory information, and an activator that triggers alerts when inventory gets low



Bi-Directional Data Flow (Batch/Real-Time)

Ingest & process

1. Millions of IoT events from multiple sources streamed in real-time
2. Contextualization data mirrored into Fabric OneLake

Analyze & transform

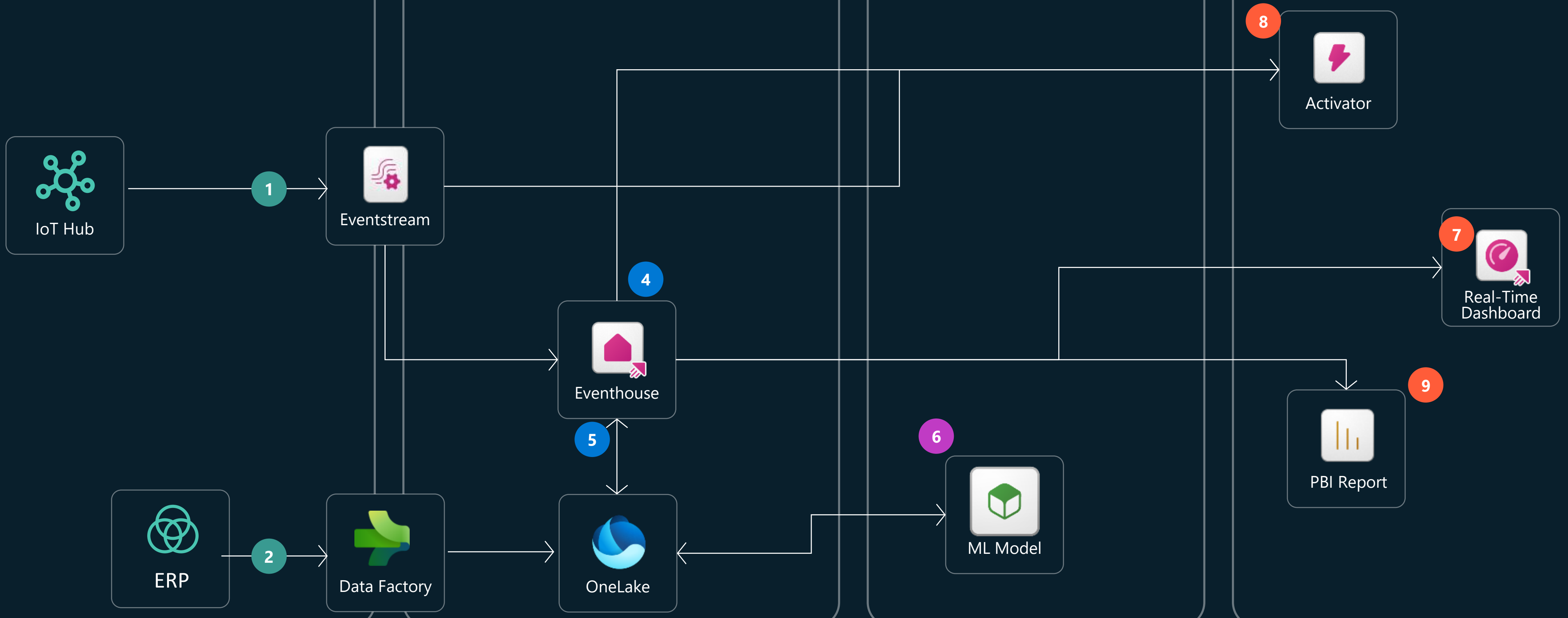
4. IoT events are enriched in motion along with data from OneLake providing clean and meaningful views
5. The aggregated tables are synced back to OneLake for serving to other Fabric workloads

Train

6. Build, train, and score ML models in real-time, persisting scored data into OneLake

Additional Workload Support

7. Real time dashboard across locations with ability to easily shift from low to high granularity
8. Enhance organizational BI reports with high granularity enriched data
9. Notify users and activate automation in response to indications from locations



Data Lakehouse Modernization

Ingest & process

1. Data is streamed in real-time from various source applications to a Fabric Eventstream
2. An Application that serves data through an API incrementally is called from a Fabric Notebook and streamed to an Eventstream
3. A Postgres database that contains reference data updated daily is loaded via Fabric Data Factory to OneLake

Analyze & transform

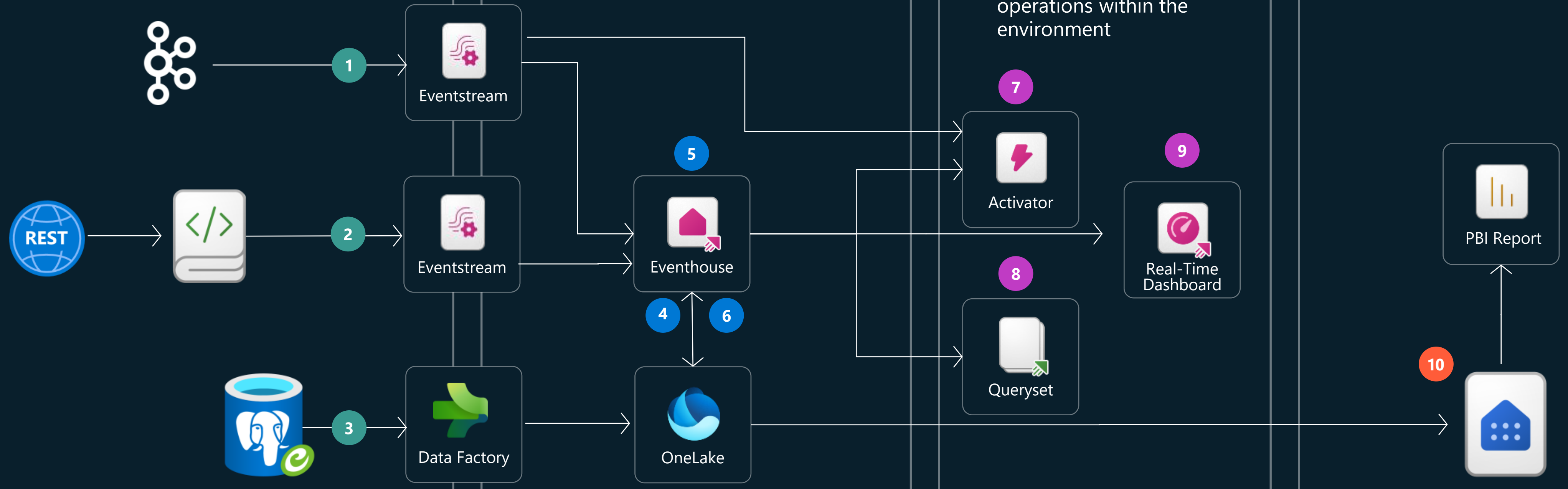
4. Data is enriched and contextualized in real-time with reference information loaded in through OneLake
5. Aggregated tables on devices are generated in real time allowing both latest and historical view on device behavior and anomalies
6. The aggregated tables are synced back to OneLake for serving to other Fabric workloads

Operationalize

7. Data Activator is used to serve alerts in real-time and near real-time to support business processes. KQL Querysets are used to dive deep and explore real-time data, as well as creating geospatial visualizations on real-time data
8. Real Time Dashboards give a current view of operations within the environment

Additional Workload Support

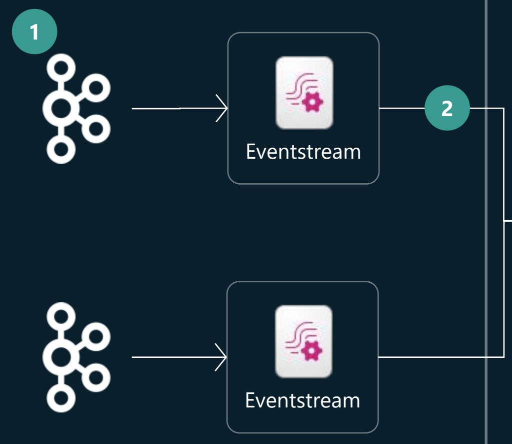
10. Fabric Data Warehouse is used to serve existing Power BI reports with minor code change



Enterprise Integration/Event Processing

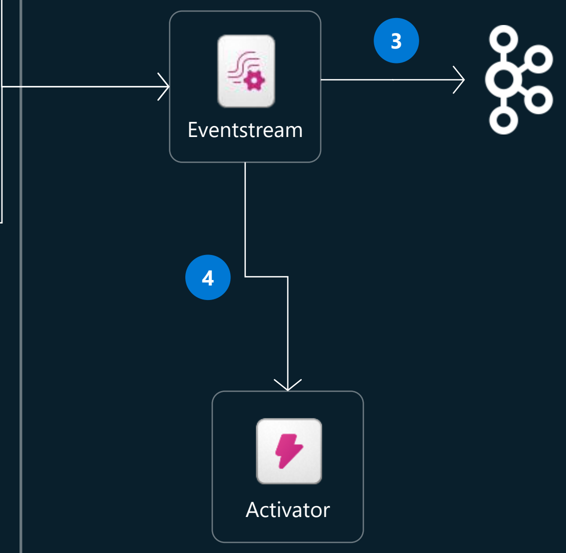
Gather

1. Kafka topics are brought into Fabric
2. Events from various topics are transformed in flight



Compare and Serve

3. Transformed events are made available as new Kafka topics
4. Activator is used to generate alerts and integrate with downstream systems



Connected Factory

Ingest & process

1. Each Plant sends data to a paired historian in the cloud
2. Contextualization data from SAP is loaded from Fabric Data Factory into OneLake

Analyze & transform

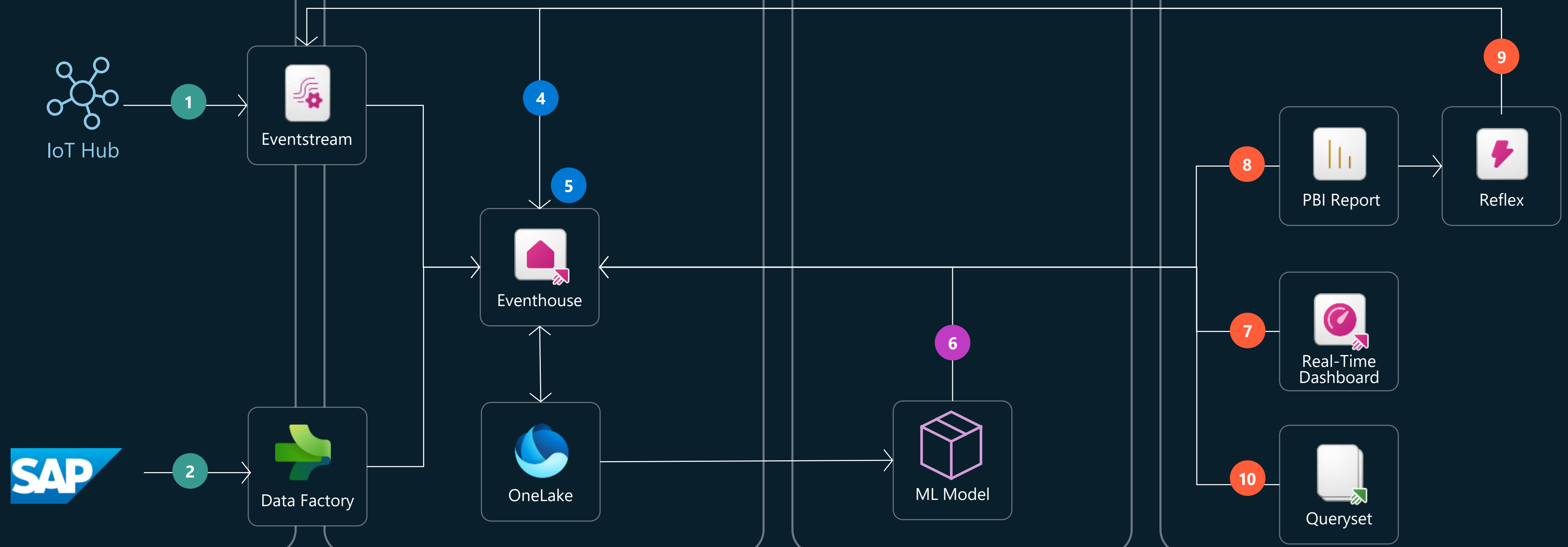
3. IIoT events are enriched in-motion with the asset meta data and asset hierarchy , providing clean and curated data for consumption.
4. Enriched data is aggregated with hourly and daily views per station and factory.

Train

6. Build, train and score ML models in real time, persisting the scored data in OneLake.

Visualize and Activate

7. Real time dashboard of all factories with ability to easily drill down from all factories view to a single asset details.
8. Rich BI reports with detailed views on factories production, efficiency and maintenance.
9. Notify on site techs in response to live indications from the factory floor.
10. Use rich KQL queries for fast, direct, ad-hoc analytics.



Ingest & process

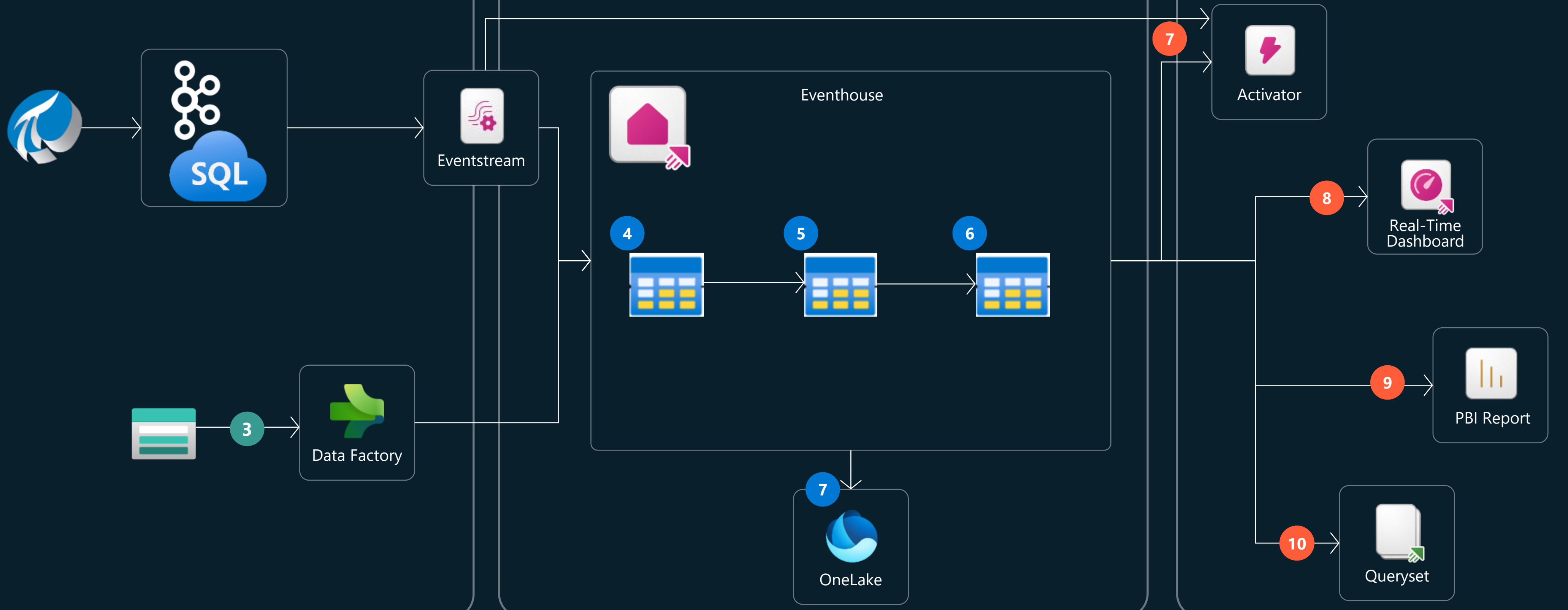
1. Millions of events are sampled every second from sensors and streamed in real time to operational systems
2. Edge Computing forwards the message to a paired cloud database or kafka topic
3. Contextual Reference data is loaded from storage into Fabric

Analyze & transform

4. Raw Data is landed directly in eventhouse
5. Data is deduped, transformed, and curate on arrival with Eventhouse Update Policies
6. Data is enriched and contextualized for reporting and dashboard views
7. Data is mirrored to OneLake for additional downstream applications

Visualize and Activate

7. Alerts are created both from contextualized data and raw events to users around key metrics
8. Real-Time Dashboards visualize operational data instantly
9. Deep Power BI reports provide historical and analytical insights
10. KQL Querysets are used to perform deep understanding and ad-hoc analysis



Data Engineering Observability

Ingest & process

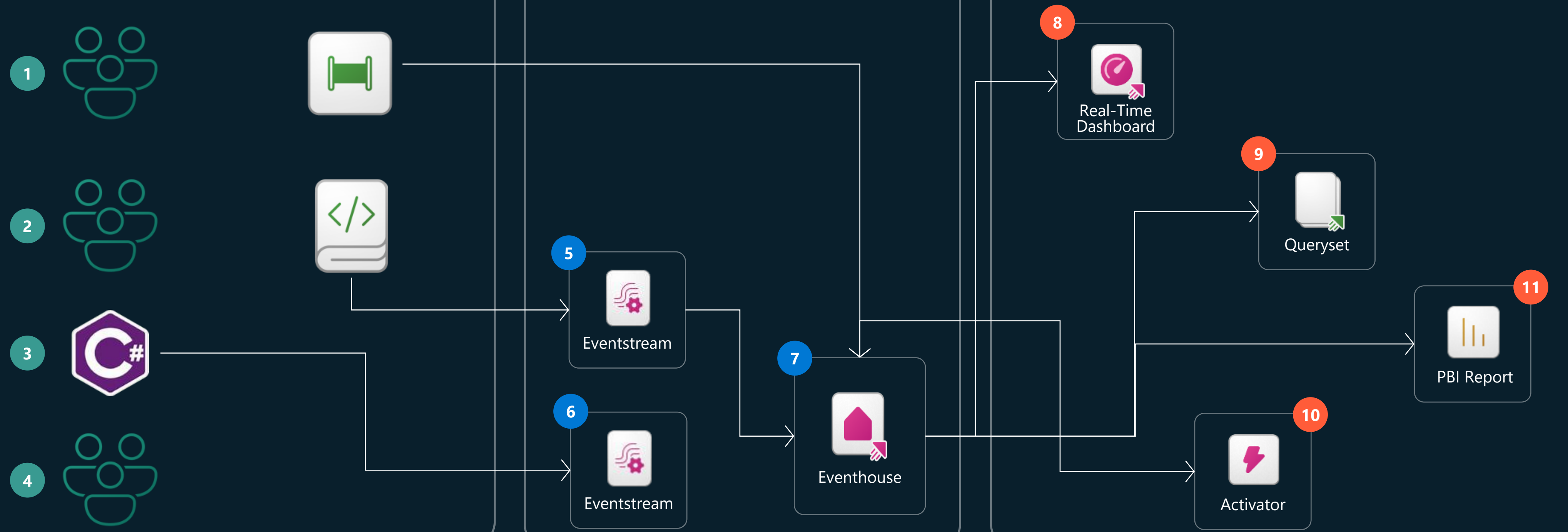
1. A business analytics team is loading data in batch from various sources to Fabric
2. A data engineering team is running a Spark notebook in Fabric
3. A custom application exists outside of Fabric that is leveraged by internal or external users
4. A centralized monitoring workspace is established to gather insights from across the organization

Analyze & transform

5. A centralized eventstream is configured that gathers notebook telemetry from across the environment
6. Another eventstream is created to listen to custom observability metrics forwarded from outside systems
7. An eventhouse is created that runs and updates this data in real-time to process and store

Visualize and Activate

8. Real Time Dashboards provide real time reporting on the current state of the environment
9. KQL Querysets are used to dive deep into application and system troubleshooting logs in real-time
10. Real time notifications push alerts when exceptions are met
11. Deep Power BI reports provide historical and analytic insights into application and cloud performance



Event Driven Arch using SQL Change Feed

Ingest & process

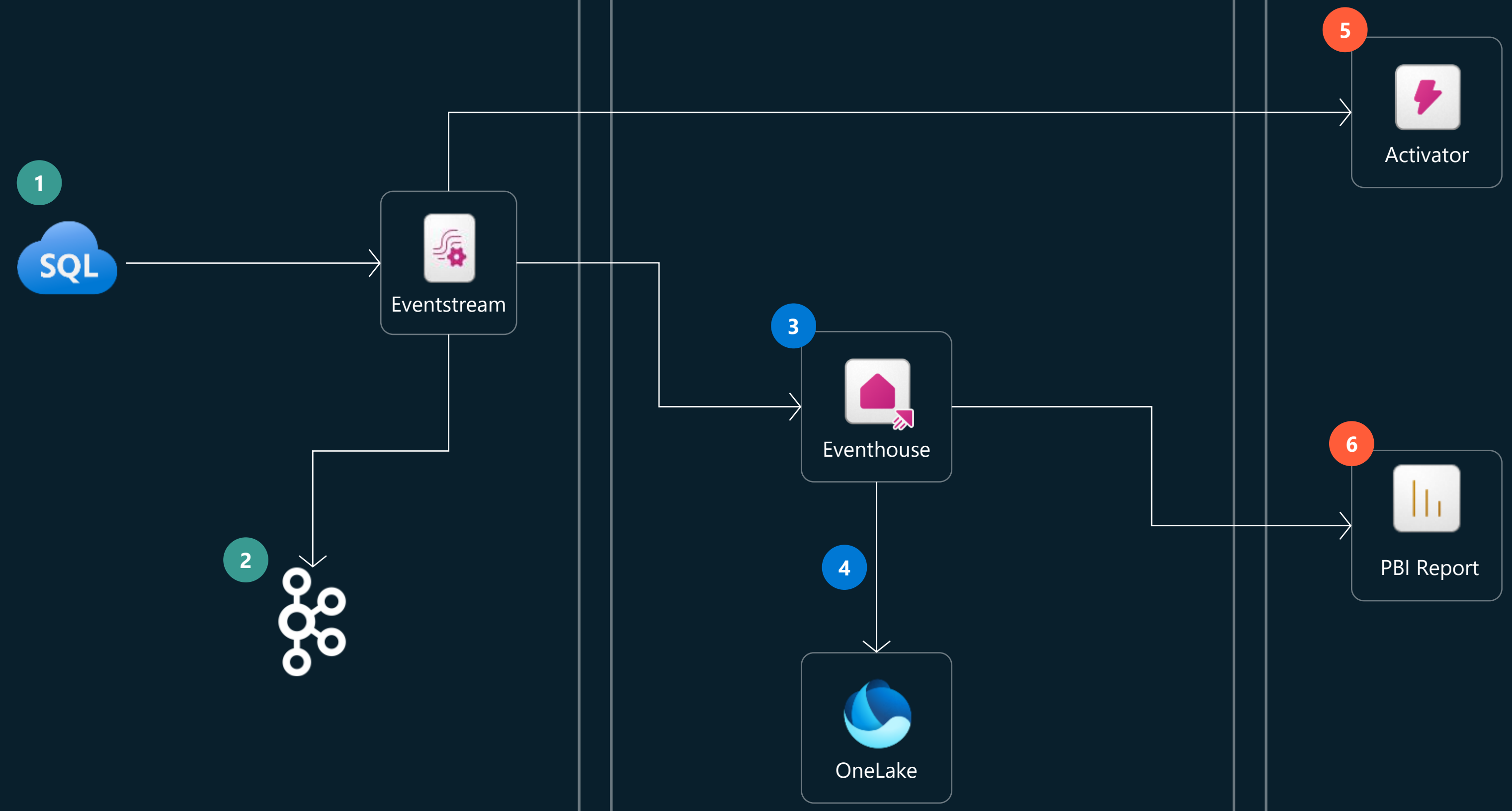
1. Azure SQL leveraged Change Event Feed to send data to an eventstream
2. Eventstreams custom endpoint is used to surface changed data as kafka topics for other enterprise scenarios

Analyze & transform

3. Data is enriched and analyzed in motion leveraging eventhouse
4. Data is replicated and sent to OneLake

Act and Visualize

5. Activators monitor data in Azure SQL directly for changes, executing a wide variety of actions based on incoming data
6. Power BI reports are built on top of Azure SQL



Machine Learning on data in real-time

Ingest & process

1. Azure SQL leverages Change Event Feed to send data to an eventstream

Analyze & transform

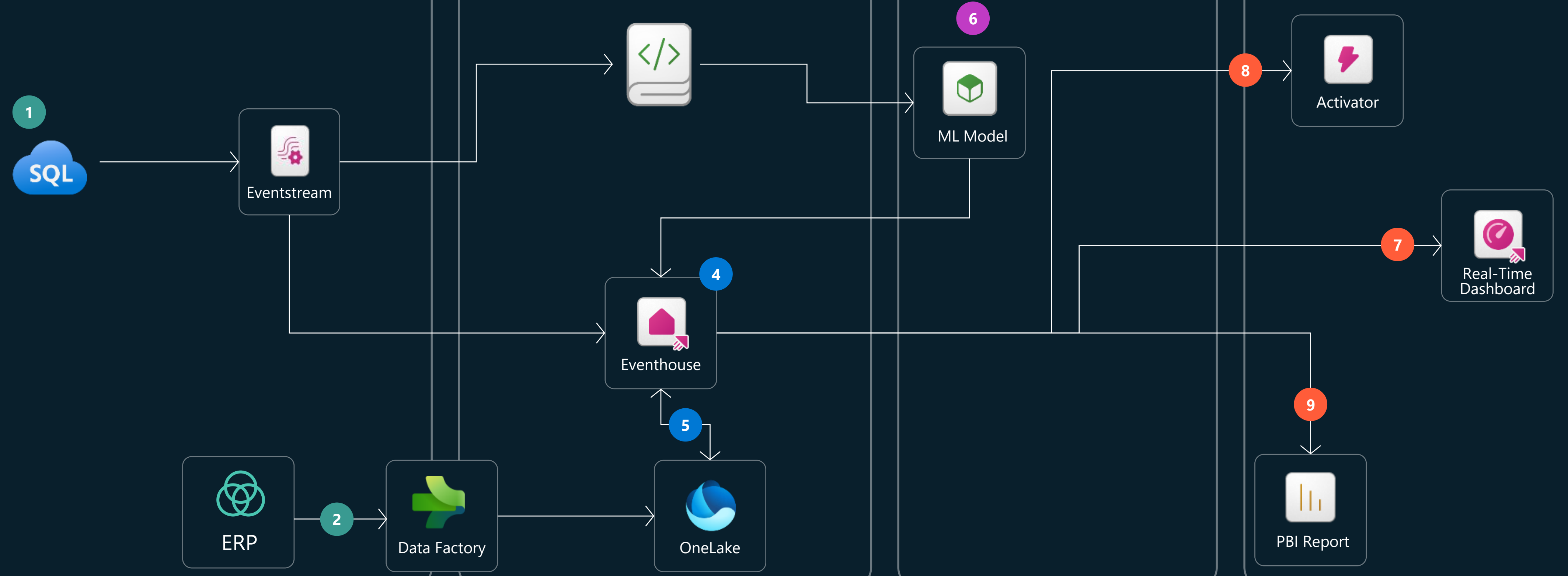
4. Data is enriched and analyzed in motion leveraging eventhouse
5. The aggregated tables are synced back to OneLake for serving to other Fabric workloads

Train & Score

6. Leveraging Fabric Notebooks, train and score ML models in real-time, persisting scored data

Additional Workload Support

7. Real time dashboard across locations with ability to easily shift from low to high granularity
8. Enhance organizational BI reports with high granularity enriched data
9. Notify users and activate automation in response to indications from locations



Factory anomaly detection hot path analytics

Ingest & process

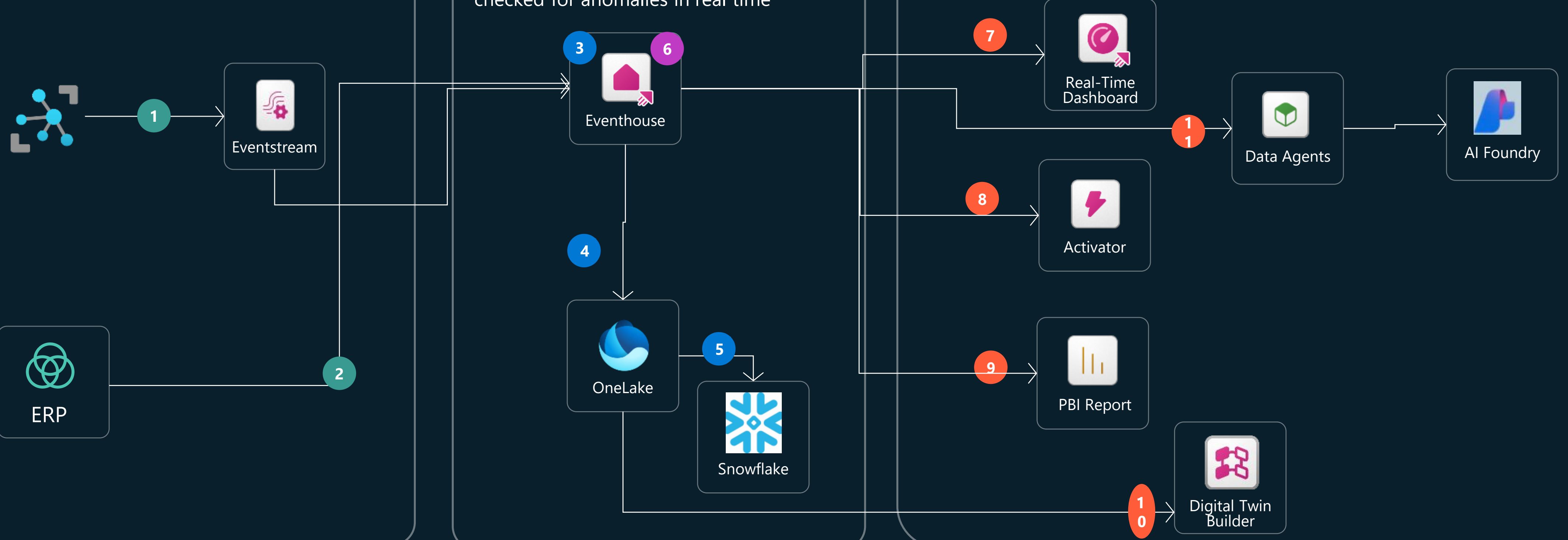
1. IoT Edge sends factory information to an existing Iot Hub implementation, which is connected to a Fabric Eventstream
2. Data from an ERP system is loaded in batch into Eventhouse

Analyze & transform

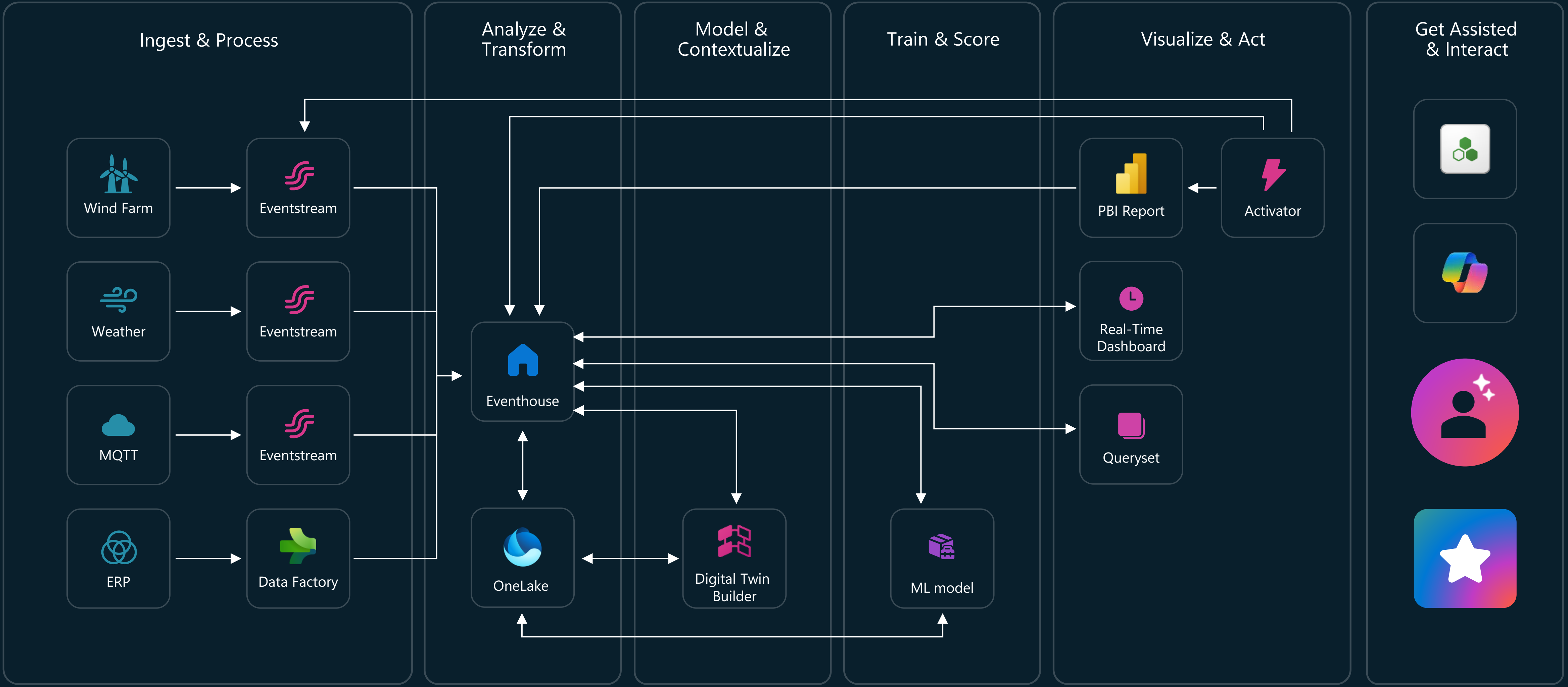
3. Data is enriched and contextualized in motion leveraging Eventhouse
4. Data is replicated to Onelake after contextualization
5. The contextualized, enriched data is copied into Snowflake for data warehouse use cases
6. Leveraging Anomaly Detection in real time intelligence, data is scored and checked for anomalies in real time

Consumption Artifacts

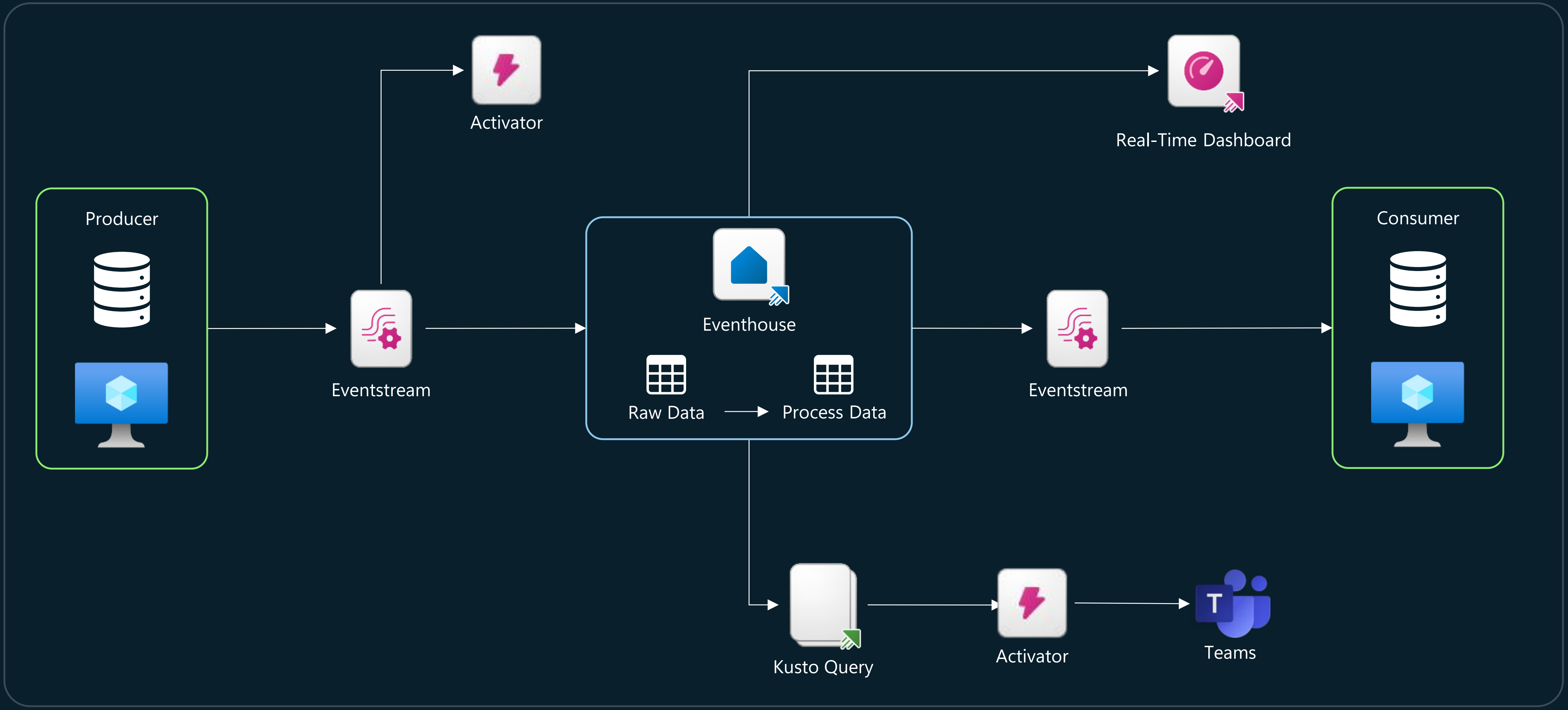
7. Real time dashboard across locations with ability to easily shift from low to high granularity
8. Notify users and activate automation in response to indications from locations
9. Create Power BI reports to seamlessly blend historical and real time reporting
10. Leverage Digital Twins to apply digital representations of relationships between data to contextualize your data for AI
11. Real Time factory line work is exposed to a data agent for conversational AI in one place. Optionally, it can be extended with AI Foundry to further ground the model or embed into Teams



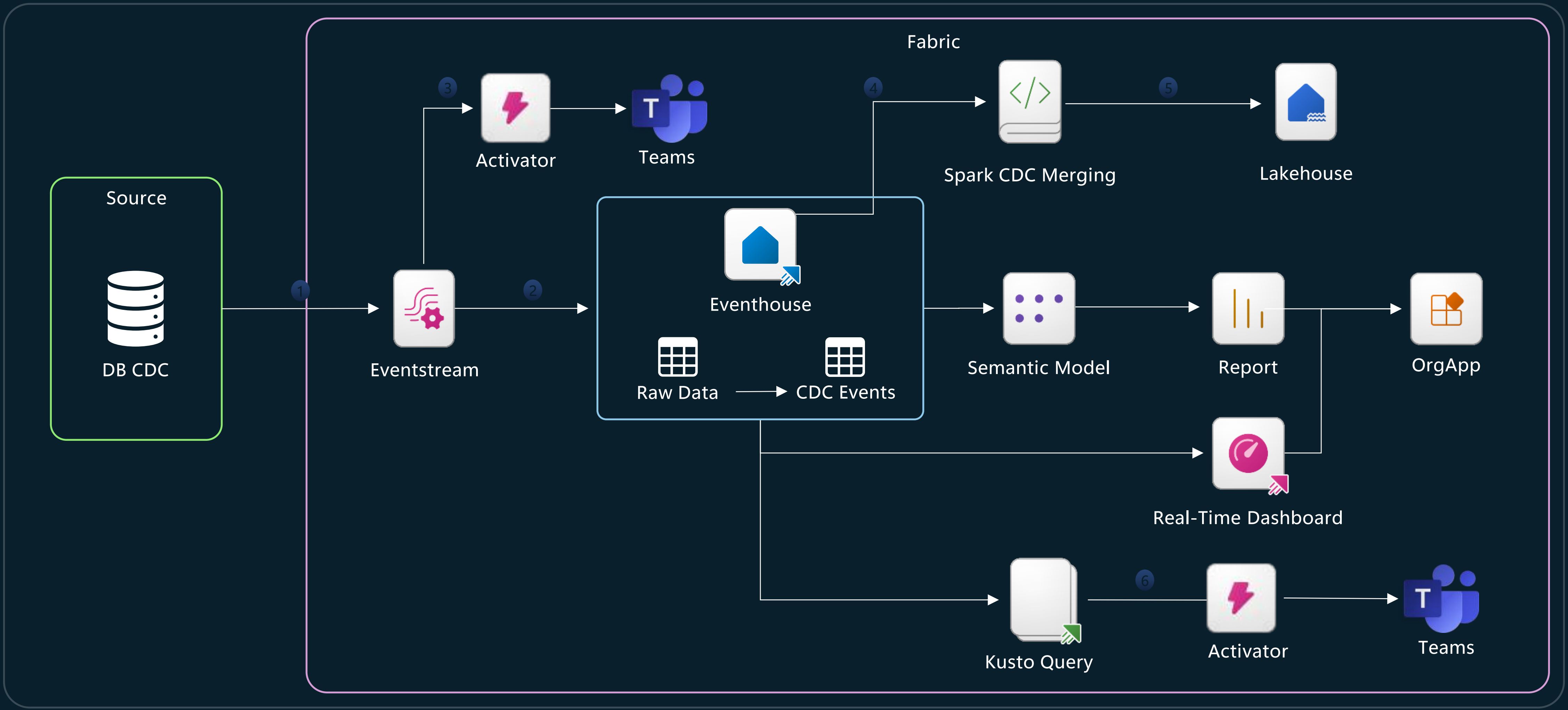
Energy Management



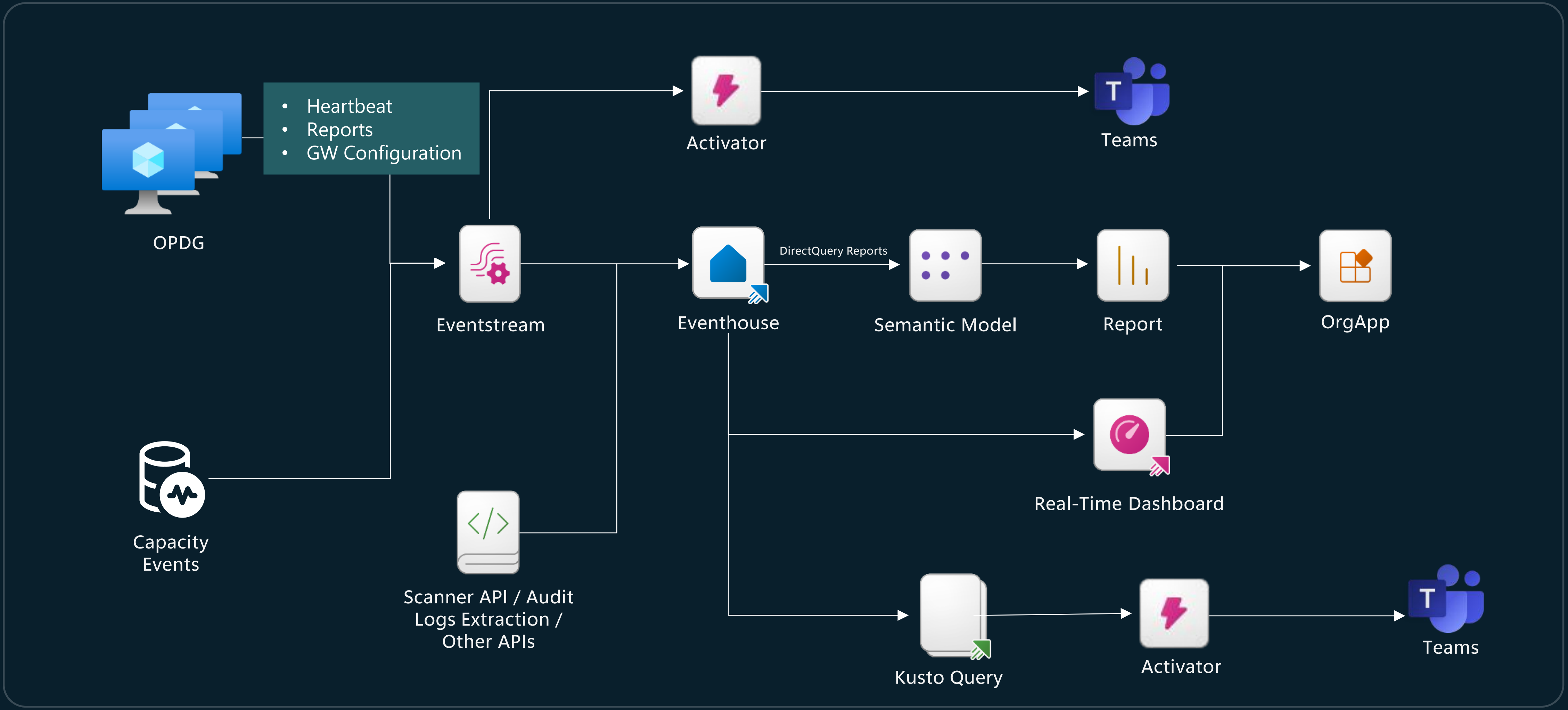
Pub-Sub Stream Processor



Database CDC Processing

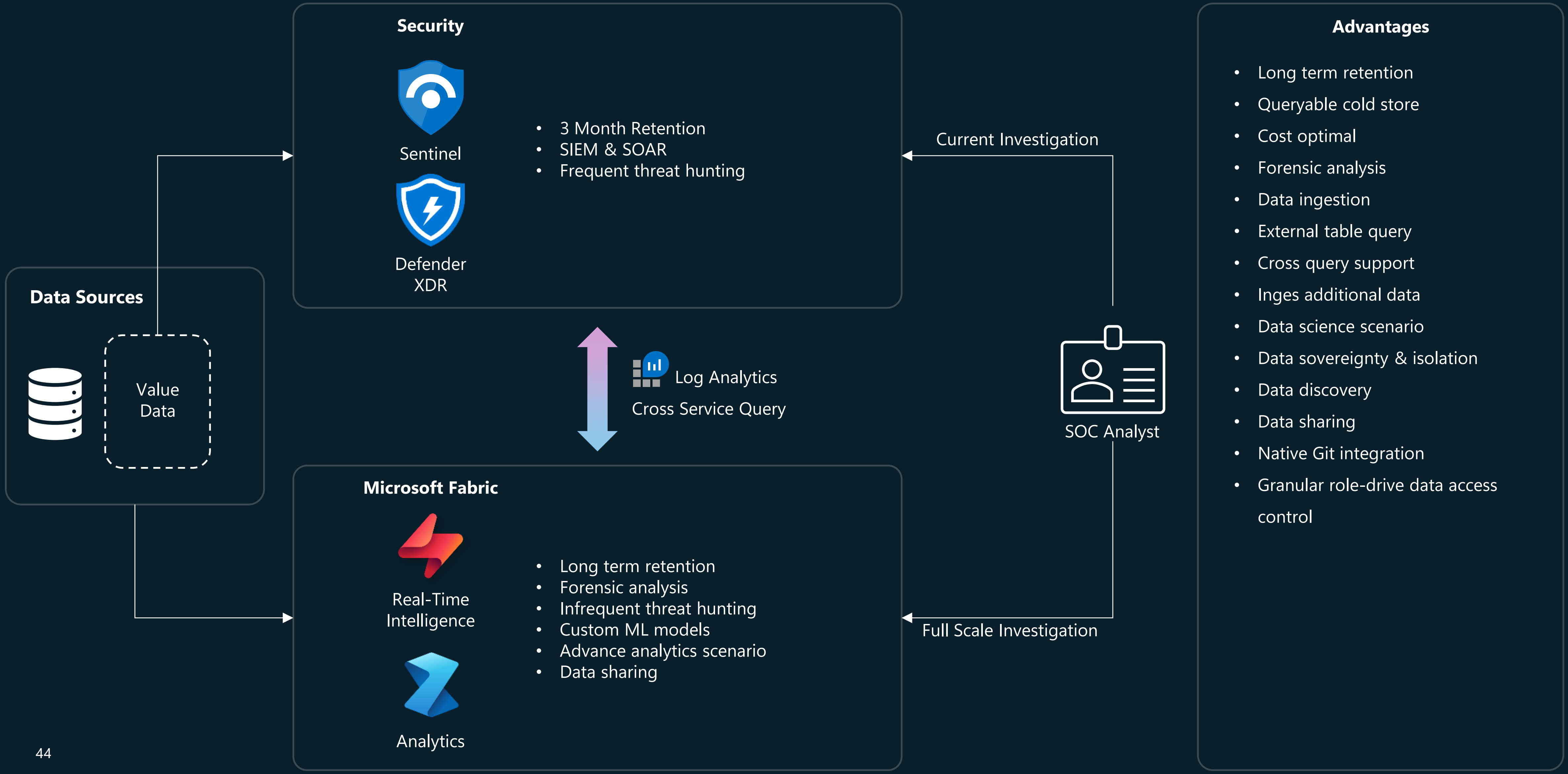


Fabric Platform Observability

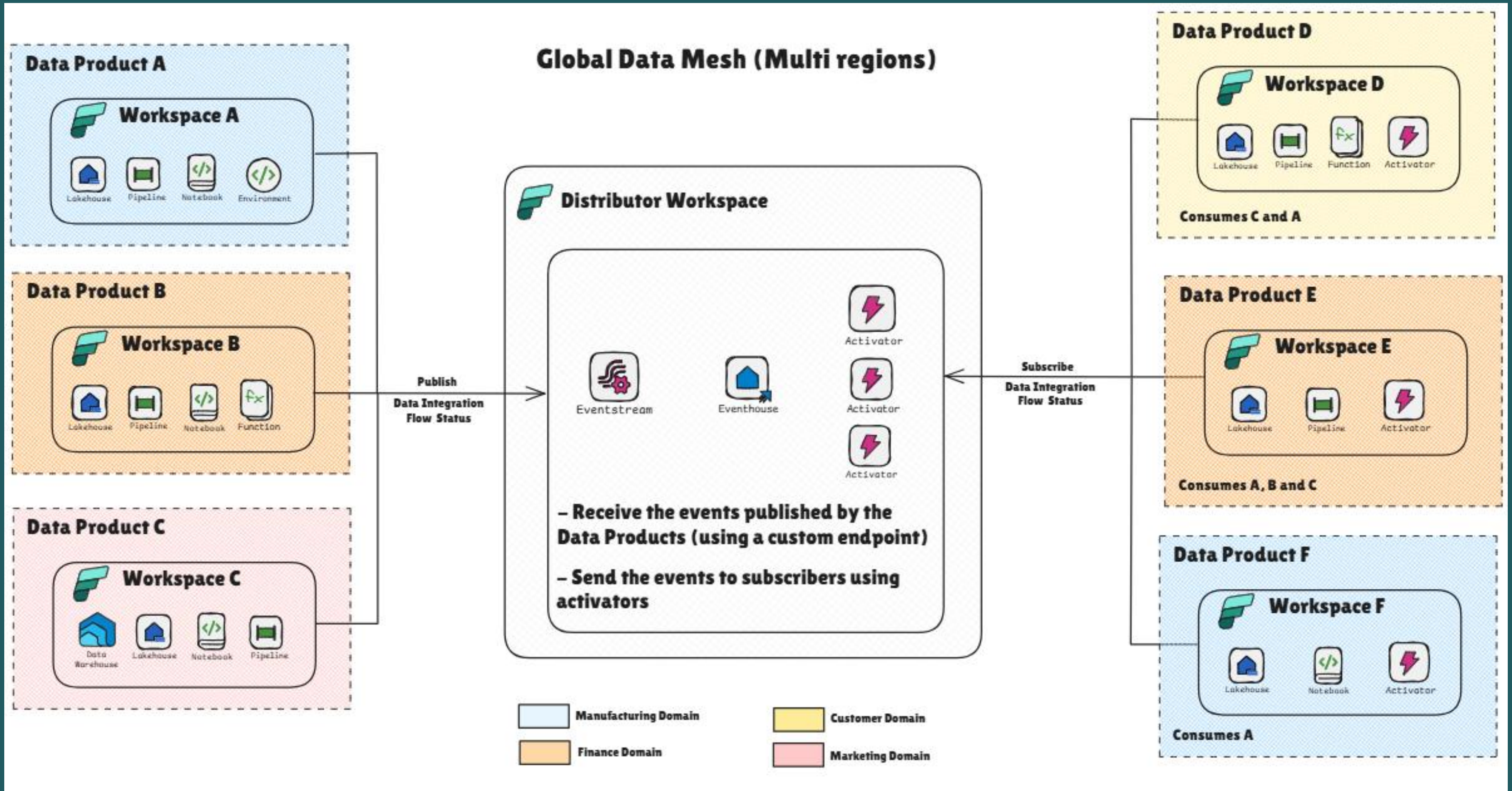


Credits: Edgar Cotte aka.ms/FabricPlatformMonitoringRTI

Sentinel Logs Offload Pattern

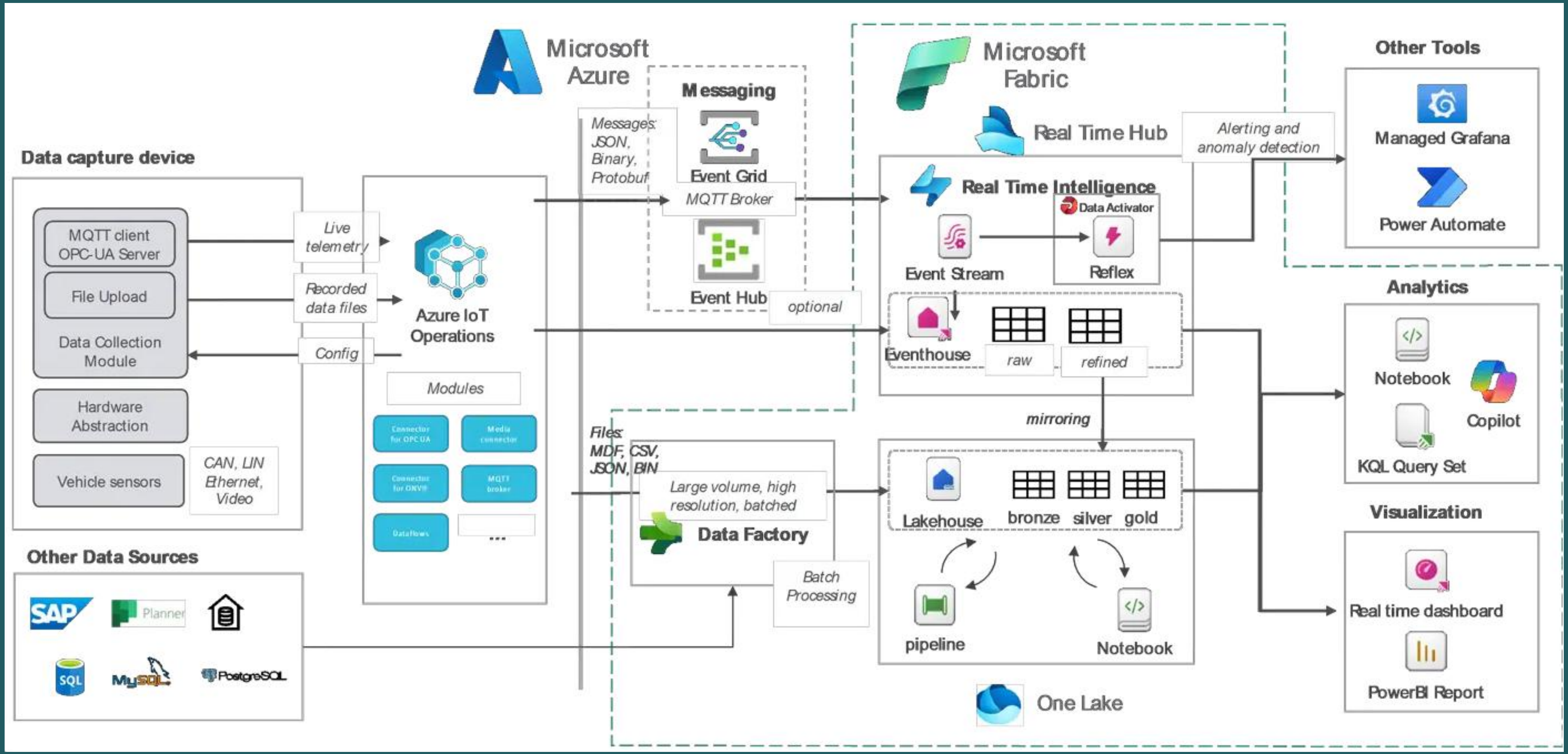


Data Product Synchronization Pattern



[(8) MS Fabric - Data Products dependency challenge | LinkedIn]

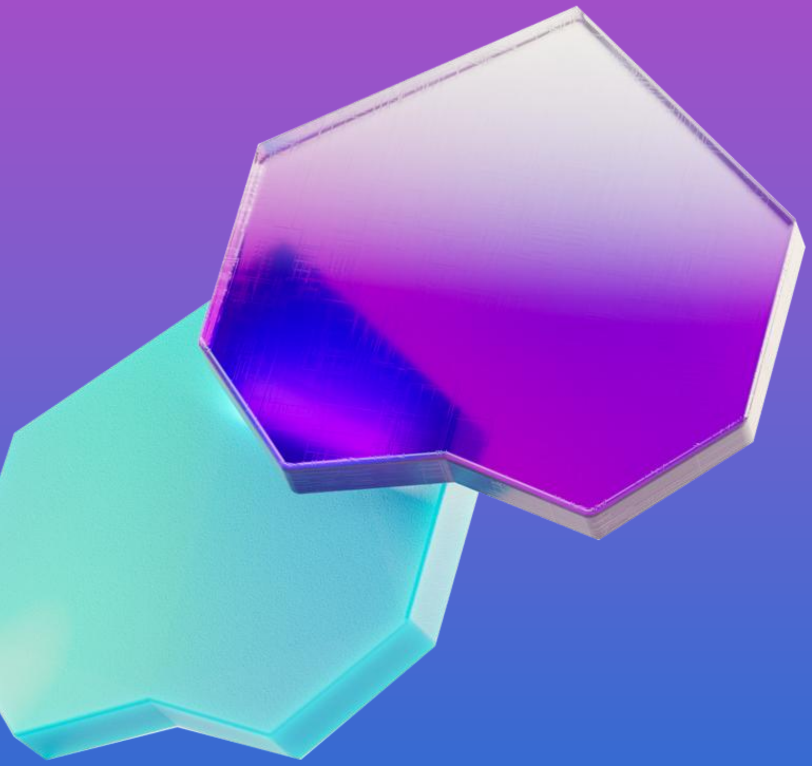
Smart Manufacturing



Credits: Florian Stein Real-Time Intelligence With Azure IoT & Microsoft Fabric

Business Discovery

Foundational Context



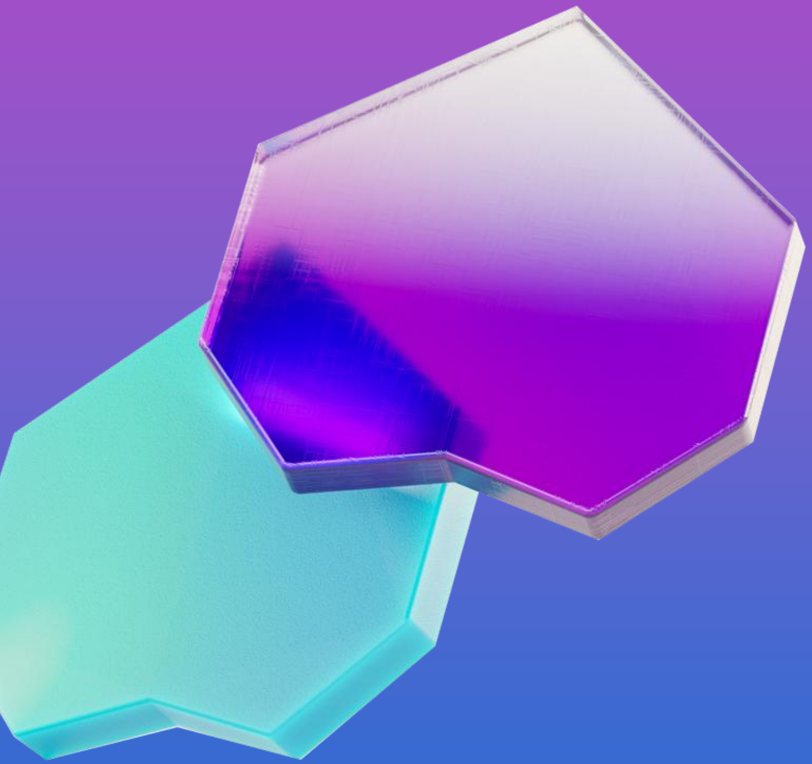
Organizational Scope

- How does your organization generate value?
- How your department contributes to the overall organization goals?
- What are the responsibilities of your department?
- How the organizational chart of your organization looks like?
- What dependencies do you maintain with the other departments?
- What do you provide to other departments as deliverables?
- What do you receive or expect from other departments?
- How do you measure the impact within your department or organization?
- How business users consume and leverage data within the organization?
- Does your organization have any experience with Real-Time data?

Team Identification

- Which stakeholders decide on the scope, the priorities and the acceptance of projects?
- Does the Data Engineering team have experience with Real-Time data?
- Who are the primary consumers of real-time insights?

Priorities and Goals

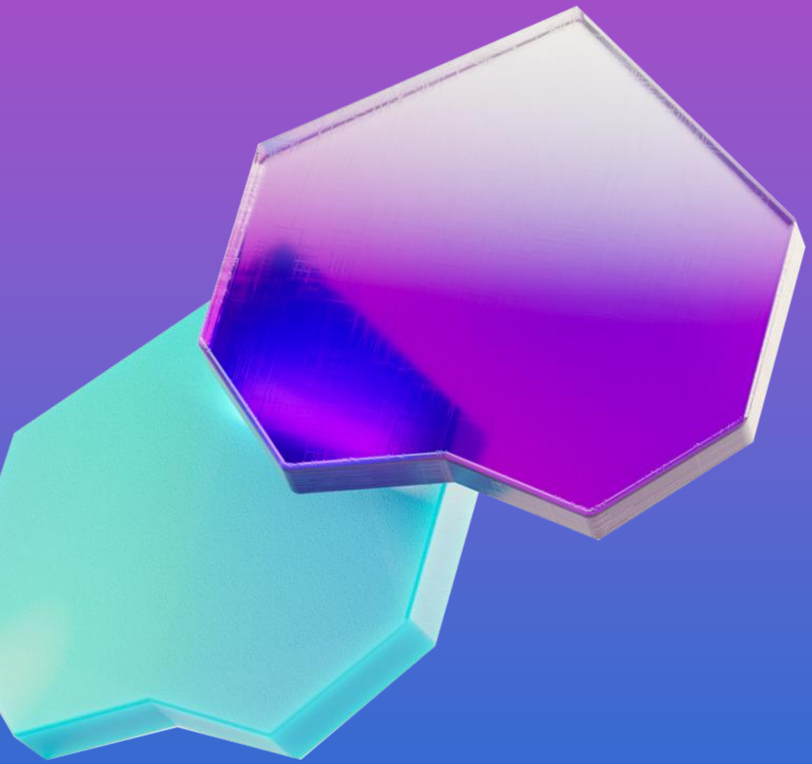


- What are your current goals and priorities?
- What are some business problems are you trying to address?
- What will be the business impact if some problems get solved?
- What data you have, or need to have, that will impact your goals and priorities?
- Which parts of your business rely on any real-time data?
- From the catalog of use cases, do you identify any of them that will contribute to reaching your goals?
- How do you envision real-time insights influencing day-to-day decisions?
- What kind of automated actions or alerts do you need?

Competitive advantage

- What competitive advantage do you expect from real-time intelligence?
- How will real-time insights provide a competitive advantage?
- Are you trying to identify new revenue streams or improve customer experience?

Priorities and Goals

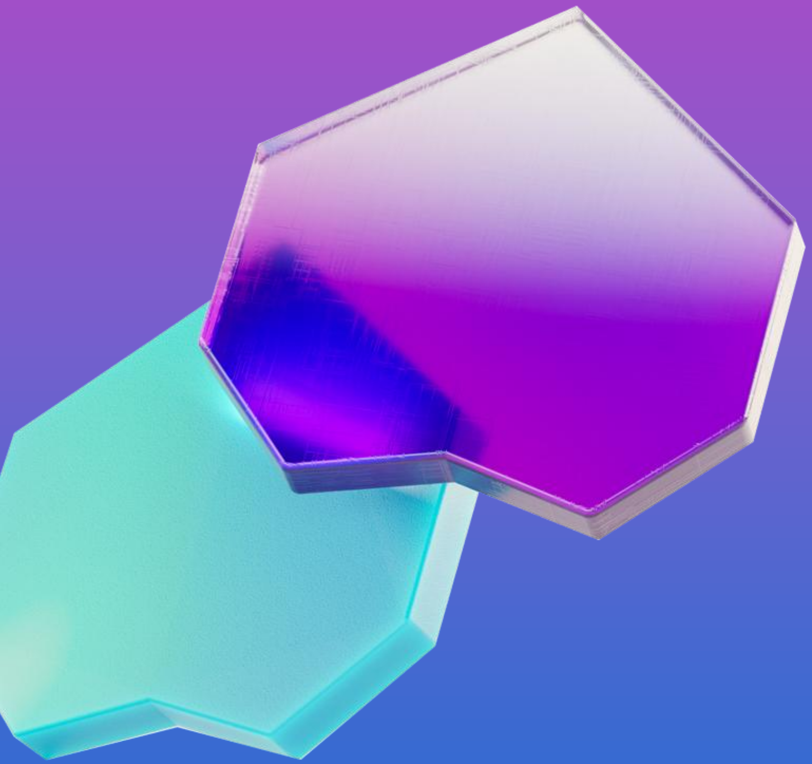


Risk mitigation

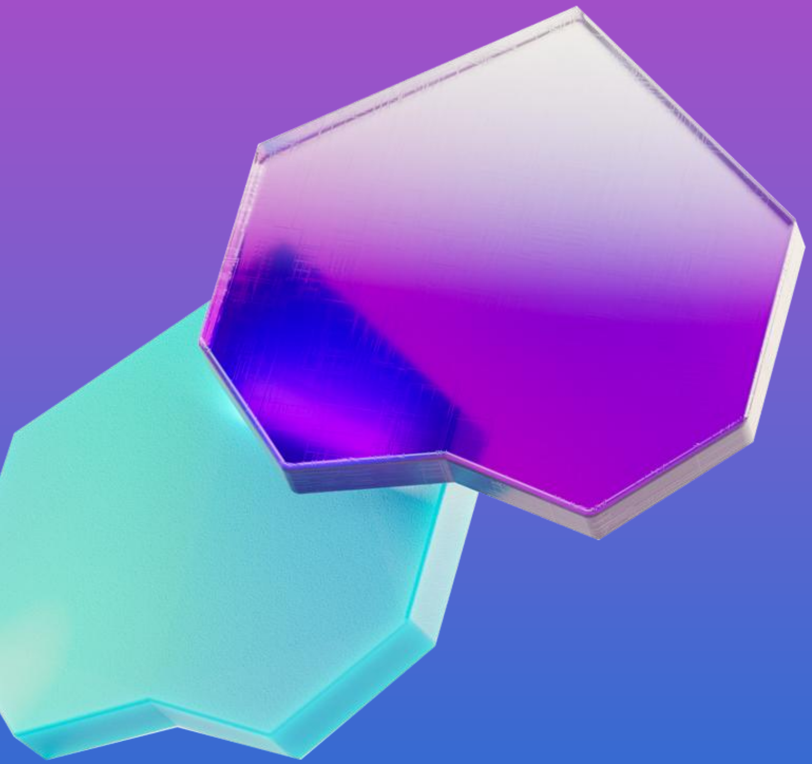
- What business risks are you hoping to mitigate?
- What happens if you do nothing and continue with your current process?
- What is the financial and operational impact of not addressing this issue?
- What are your top business outcomes for the next 6–12 months that could benefit from real-time insights?
- Which decisions currently suffer from latency in data availability?
- Are there specific KPIs or metrics you wish you could monitor in real time?
- How do you envision democratizing access to insights across departments?
- What are the key strategic business initiatives driving the need for real-time intelligence?
- How will a real-time intelligence capability enable new products, services, or business models?

Pain Points

- What part of your current business process is most frustrating or inefficient?
- How long does it currently take to get insights from your data?
- What are the consequences of these delays?
- What existing tools or processes are you using to handle data, and what are their limitations?
- Where are your teams spending significant time and effort: on data collection, processing, or reporting?
- As your business grows, what are your biggest concerns regarding data volume, velocity, and variety?
- What opportunities are you missing out on because you lack access to real-time information?
- What are the current frustrations or limitations with existing systems or reporting methods?
- What business decisions are currently being impacted by delayed data?
- Where do you feel the most pain from not having data in real time?
- Are you missing critical events or issues because you can't monitor data streams as they happen? How do you currently get alerted to issues in the environment?
- Are your teams making decisions based on intuition or outdated data because current insights aren't fresh enough?



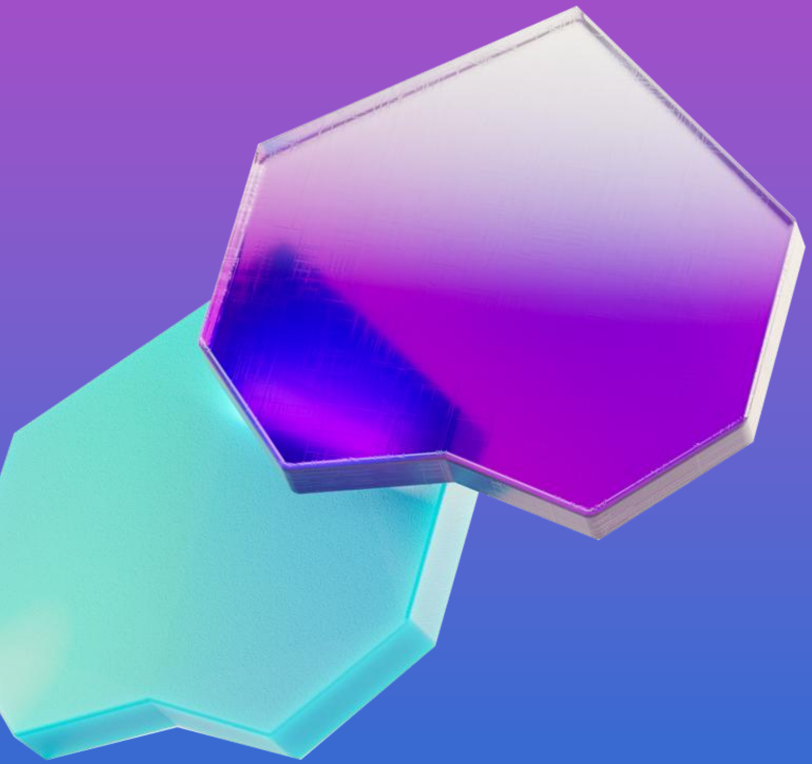
Pain Points



Cost and Complexity

- What are the current challenges related to the cost and complexity of your existing data infrastructure?
- What operational bottlenecks could be resolved with faster data insights?
- Are there manual interventions today that could be replaced with automated responses?
- What is the cost of delayed decision-making in your workflows?
- How do you currently handle alerts or exceptions in your operations?
- Are there any compliance or governance challenges that real-time data could help address?
- What are the current pain points in integrating streaming data sources?
- Are there scalability or performance concerns with your existing architecture?
- Are there compliance or SLA risks due to delayed data?

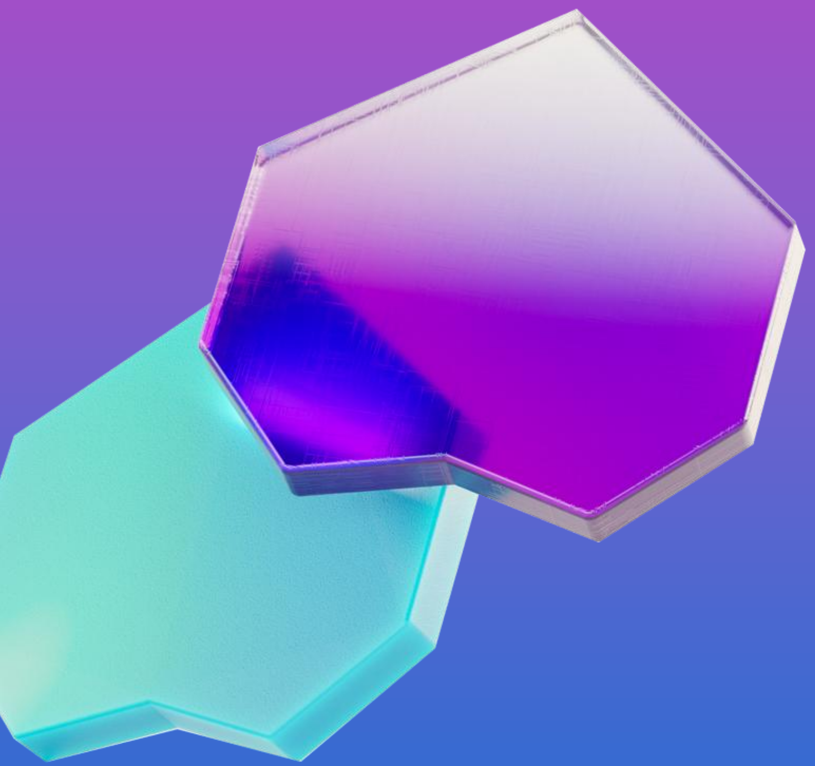
Specific Project



- Will the project be used by one or multiple departments?
- What does success look like for this project?
- How will you measure the ROI of the solution?
- How is success for the project measured?
- What are the KPI, metrics, or operations improvements you expect to see after the solution is implemented?

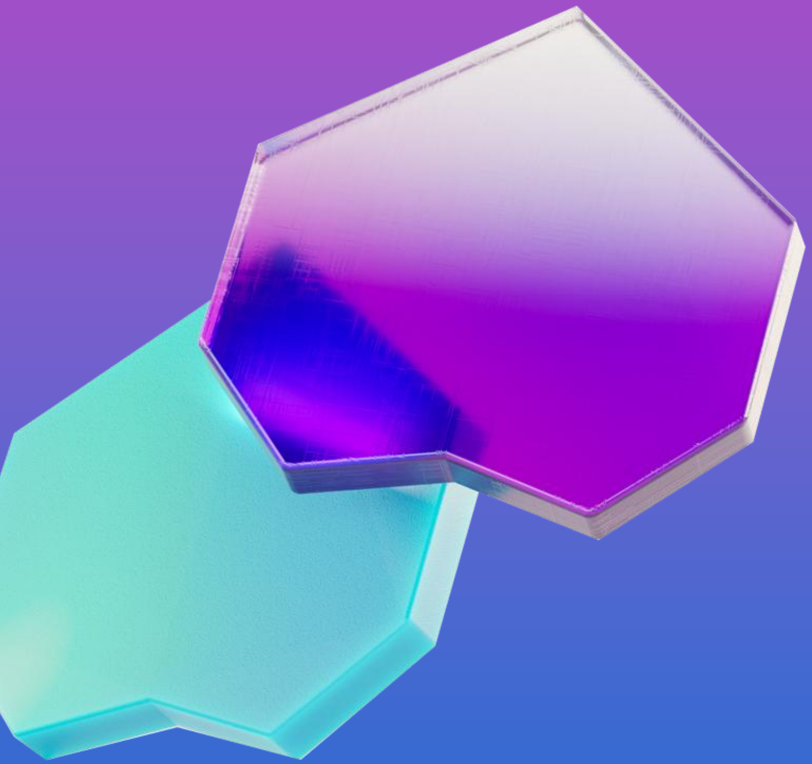
Technical Discovery

Current Systems



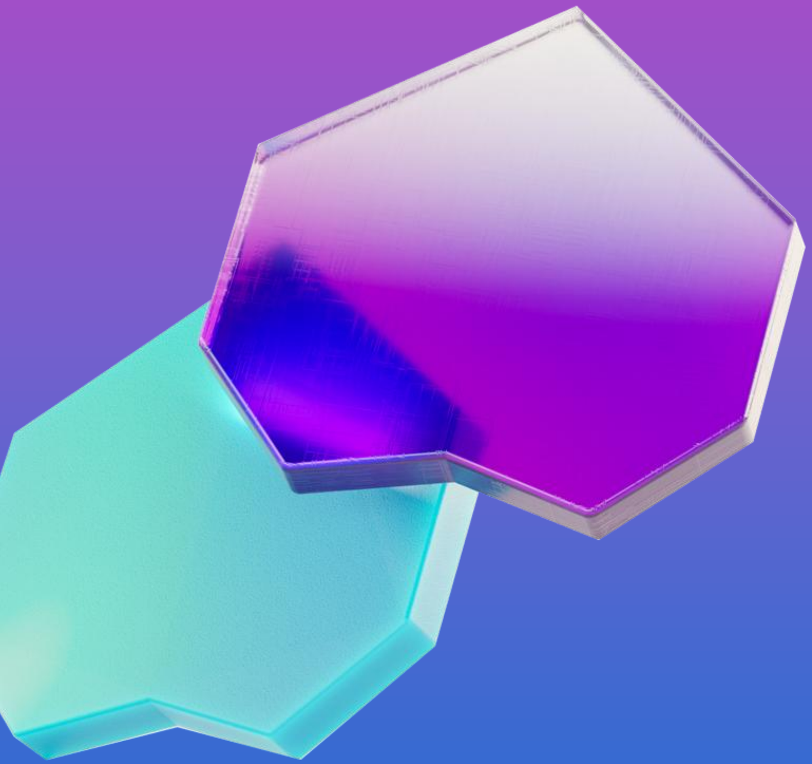
- What ERP, CRM, or line-of-business systems are in use?
- Are there any industry-specific platforms (e.g., SCADA, POS, IoT)?
- Do you use event brokers (Kafka, Azure Event Hubs, RabbitMQ)?
- Are APIs available for key systems?
- Are there legacy systems that pose challenges for real-time integration?
- Which tools are used for reporting and dashboards (Power BI, Tableau)?
- Are teams familiar with streaming technologies or KQL?
- Do you have internal resources for real-time monitoring?

Data Sources



- What are your primary data sources?
- For each data source you have, what is the:
  - Volume? How much data is being generated in average per hour or per day?
  - What is the expected growth rate over time?
  - Velocity? How fast is the data flowing? Are there peak times or continuous streams?
  - Variety? What are the different formats and structures of the data (e.g., JSON, text logs, relational database records)?
- Are there any high-value data streams currently underutilized due to latency or integration challenges?
- How do you manage data ingestion from external platforms like AWS, Snowflake, or Kafka?
- What is your current approach to data quality and lineage across these sources?
- Where does your real-time data originate? Is it from application logs, transaction systems, external APIs, IoT devices, or other sources?
- What is the format of your incoming data? Is it structured, semi-structured (e.g., JSON), or unstructured?
- What processes are currently in place, if any, for ingesting this data? For example, are you using Azure Data Factory, or are these processes manual?

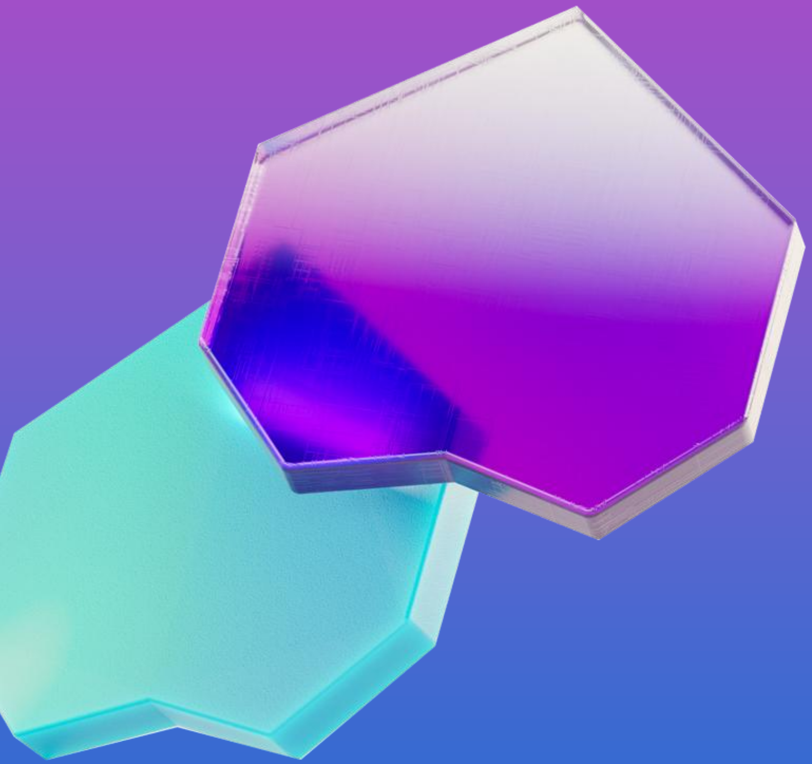
Data Sources



- Are there clear contracts or schemas defined for the data from each source, or will we need to perform discovery and validation?
- What methods and protocols are used to access the data? For example, are they using APIs, message queues (like Kafka), or database connectors?
- What historical data needs to be integrated with the real-time streams to provide complete insights?
- What are all the potential sources for the data you need to analyze in real time?

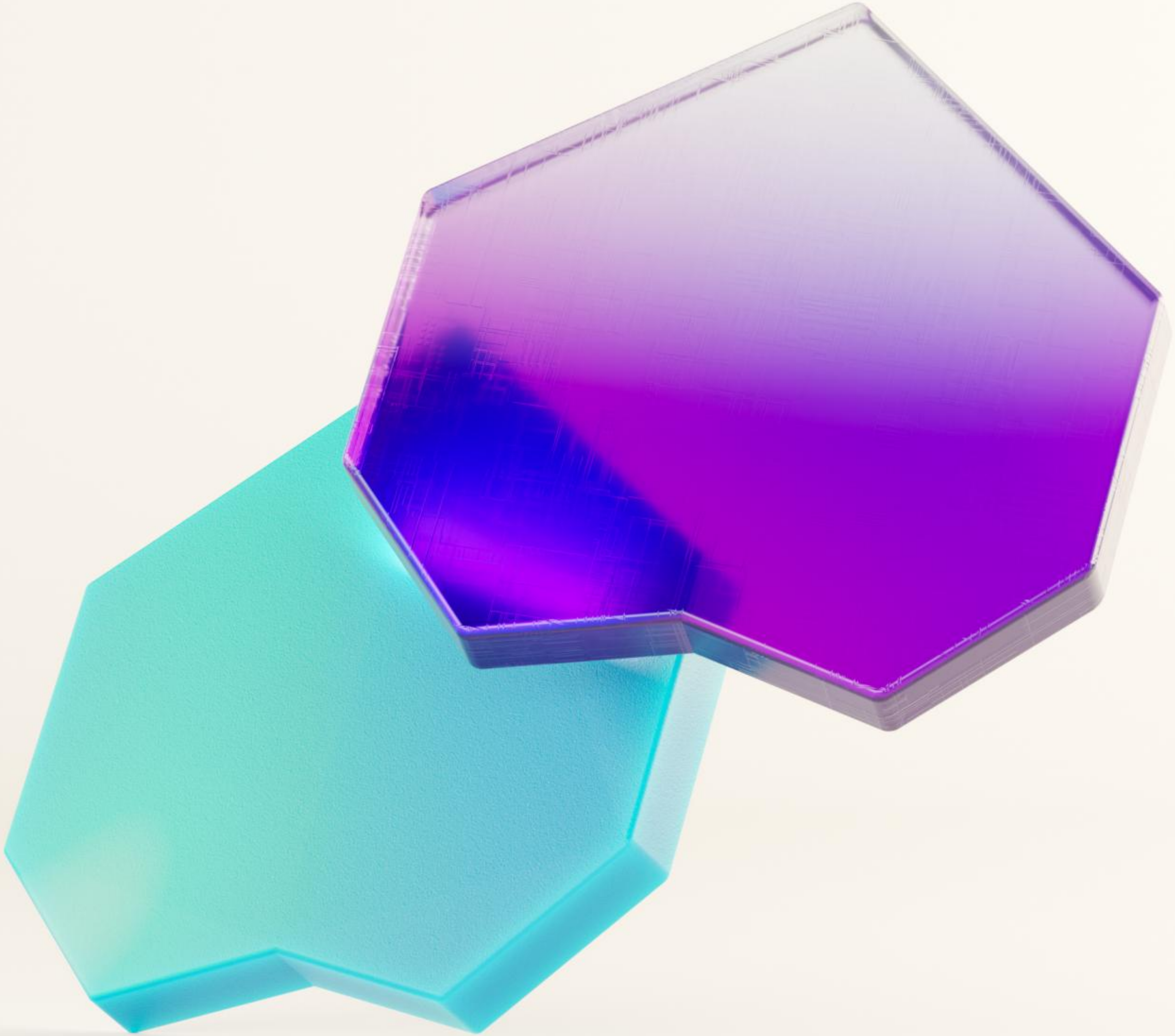
  Examples: financial transaction systems, server logs, social media feeds, IoT devices, website clickstreams and mobile applications.
- Where are these data sources located? Are they on-premises, in another cloud, or already within Azure?
- What are the security and access requirements for these data sources?
- Are there compliance or governance regulations that need to be followed?

Technical Strategy



- What governance or compliance challenges could real-time data help address?
- How do you plan to scale data access securely across business units?
- What customer experiences or outcomes are you aiming to improve with real-time data?
- What business risks could be mitigated with predictive or streaming analytics?
- Which processes would benefit from real-time monitoring or automation?
- What are the most critical events or triggers in your business that require immediate action?
- What challenges do you face in correlating streaming data with historical trends?
- Are there recurring reports or dashboards that could be enhanced with live data?

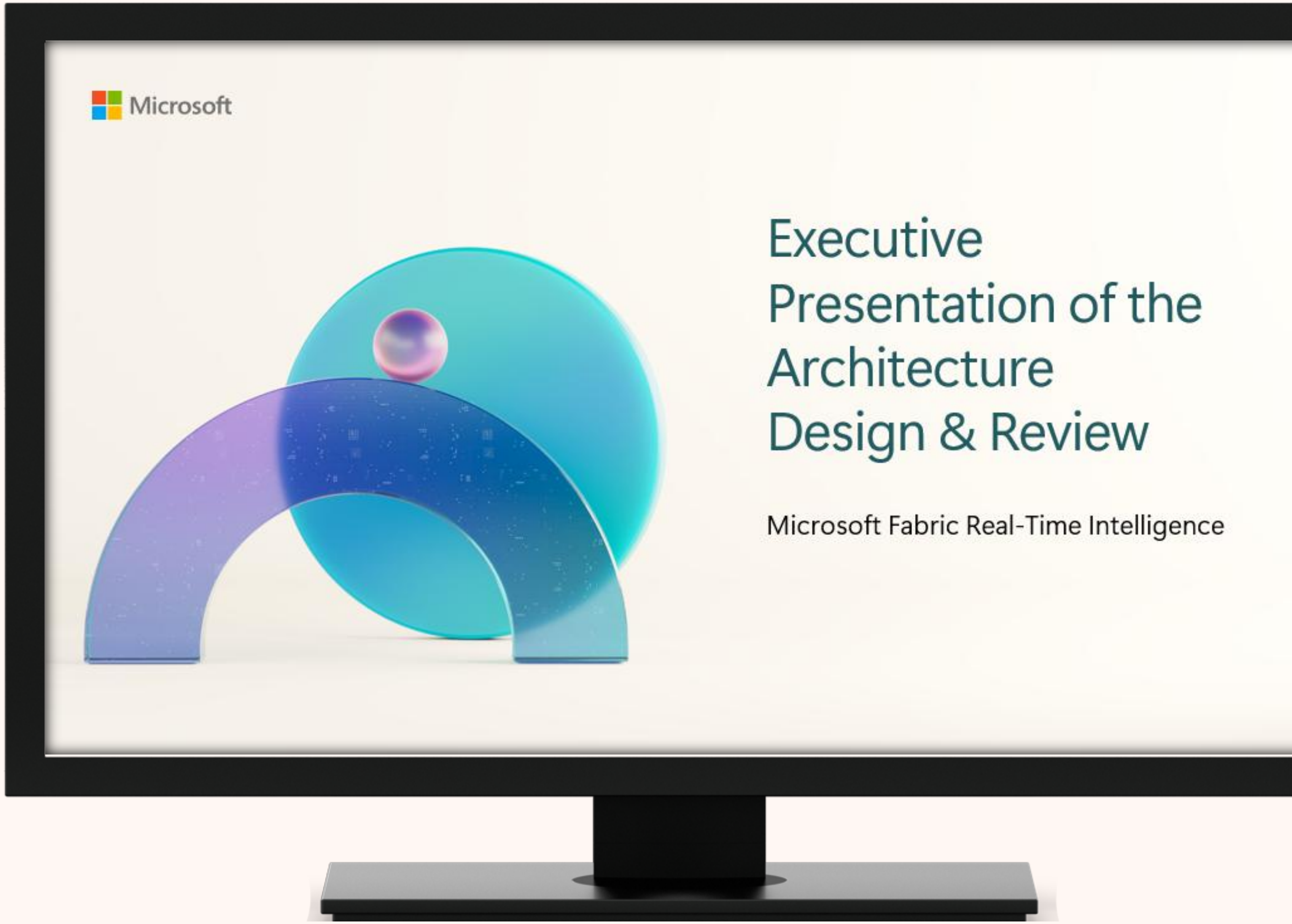
Analysis of Needs



Planning Day

Executive Presentation

- Findings
- Action Plan
- Best Practices
- Proposed Architecture
- Offerings
- Engagement Report



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